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Abken et al.

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(54) **CHIMERIC ANTIGEN RECEPTOR AND ITS USE**

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Andreas Hombach, Brühl (DE)

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(30) **Foreign Application Priority Data**

Jul. 16, 2014 (EP) 14177278

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A61K 39/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
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(Continued)

(58) **Field of Classification Search**
CPC C07K 16/2803
See application file for complete search history.

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Primary Examiner — Mark L Shibuya

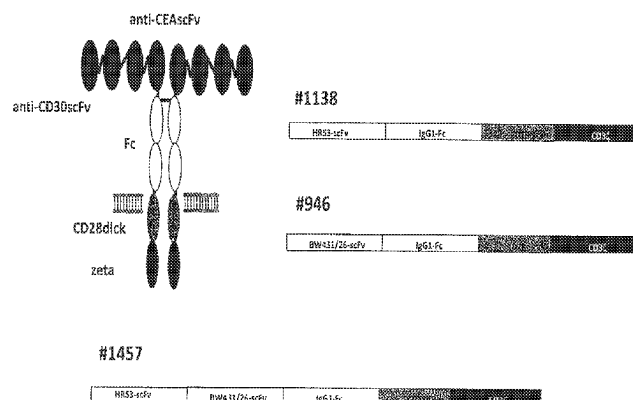
(74) Attorney, Agent, or Firm — WCF IP

(57) **ABSTRACT**

In a first aspect, the present invention relates to a recombinant polypeptide containing a domain comprising at least two antibody units whereby the first antibody unit is an anti-CD30 single chain antibody unit while the second antibody unit is an antibody unit being specific for an antigen present on the surface of a pre-determined target cell. In particular, the present invention relates to a recombinant polypeptide containing at least the following domains starting from the N-terminus to the C-terminus: a first domain containing an anti-CD30 single chain antibody unit, in particular, HRS3 scFv of SEQ ID No. 2 or homologs thereof having at least 70% identity with SEQ ID No. 2 binding specifically to CD30, and an antibody unit said antibody unit being specific for an antigen present on the surface of a predetermined target cell, in particular, being specific for a tumor-associated antigen; optionally a spacer domain; a trans-membrane domain; and a cytoplasmatic signalling domain. In a further aspect, the present invention relates to a nucleic acid molecule encoding the polypeptide according to the present invention, as well as vectors and cells containing the same. Moreover, lymphocytes are provided, in particular T-cells like CD8⁺ or a CD4⁺ T-cell expressing on its surface chimeric antigen receptors containing an anti-CD30 single chain antibody unit and an antibody unit whereby said antibody unit being specific for an antigen

(Continued)

Modularly composed CAR with dual CEA/CD30 specificity



present on the surface of a predetermined target cell. Immune cells modified with the polypeptide show improved functions, in particular in the treatment of cancer, in particular CD30⁺ cancer. That is, the cells are for use in adapted cell therapy for treating cancer in a subject in need thereof.

13 Claims, 6 Drawing Sheets

Specification includes a Sequence Listing.

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C07K 14/725 (2006.01)

C07K 14/735 (2006.01)

C07K 16/30 (2006.01)

(52) **U.S. Cl.**

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2039/505 (2013.01); **C07K 2317/24** (2013.01);
C07K 2317/622 (2013.01); **C07K 2319/00**
(2013.01); **C07K 2319/02** (2013.01); **C07K**
2319/03 (2013.01)

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Modularly composed CAR with dual CEA/CD30 specificity

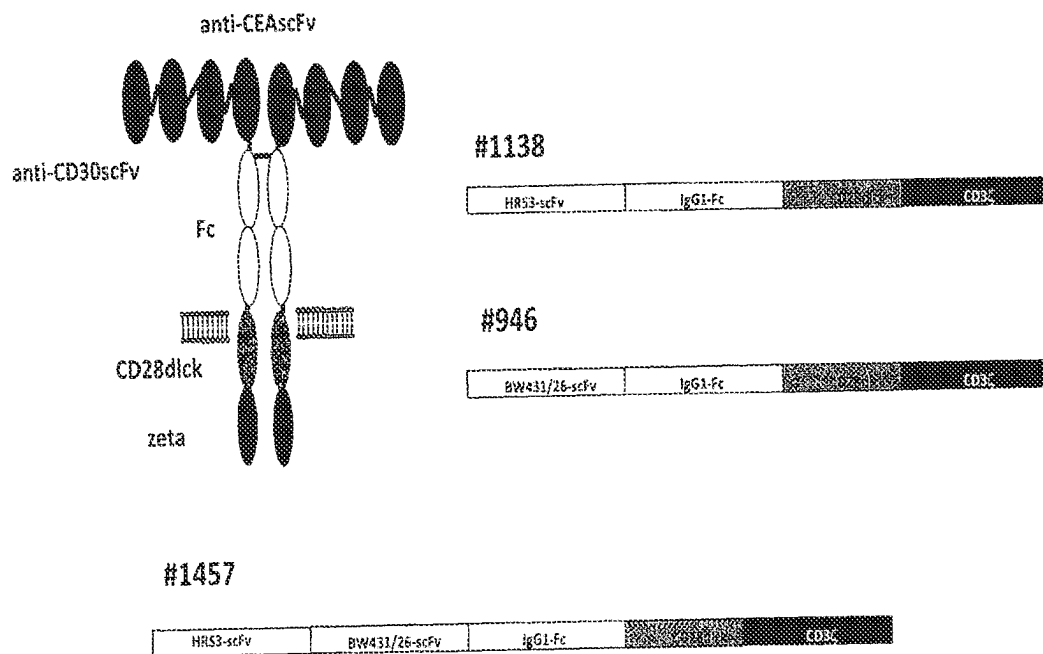


FIGURE 1

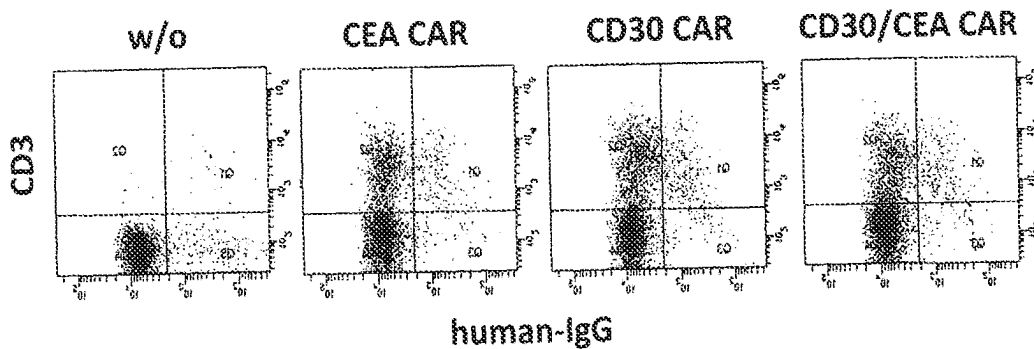


FIGURE 2

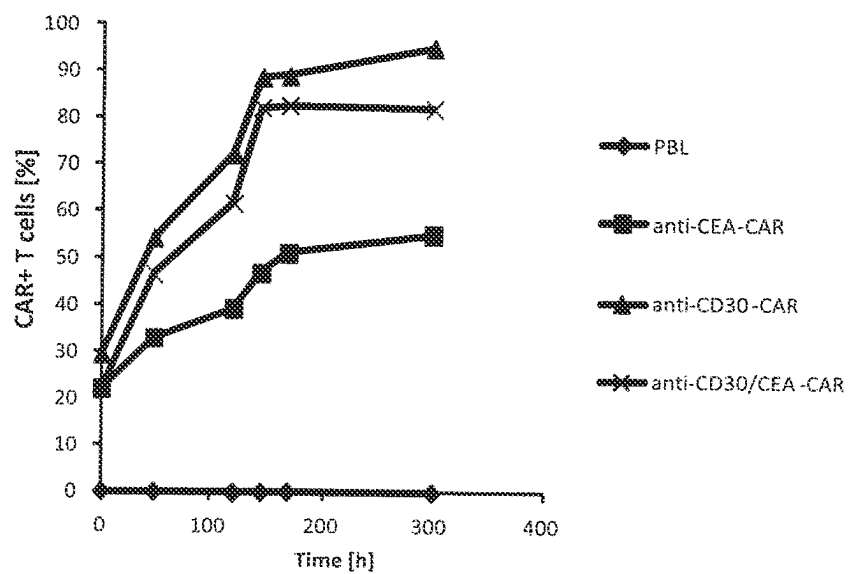


FIGURE 3

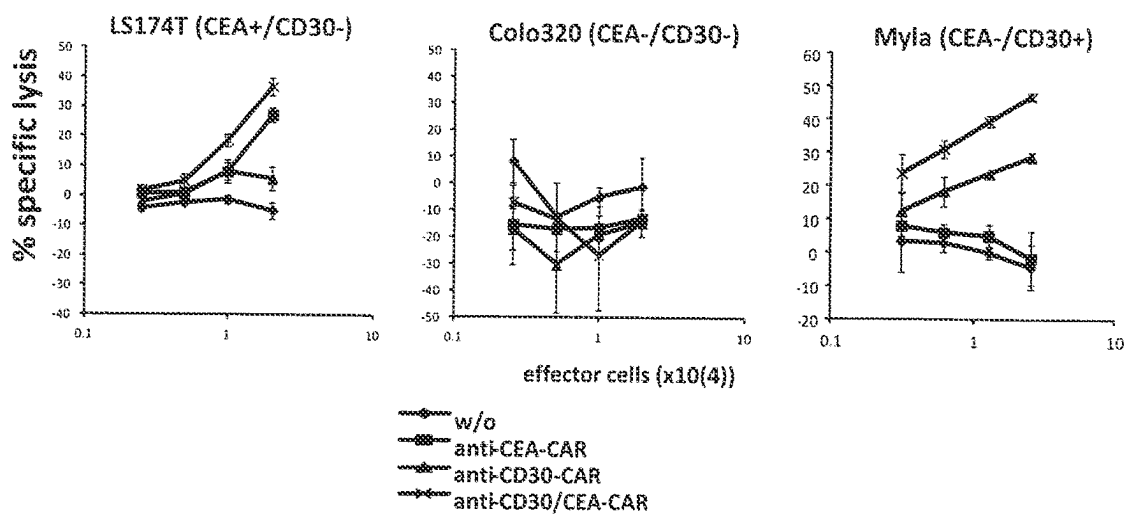


FIGURE 4

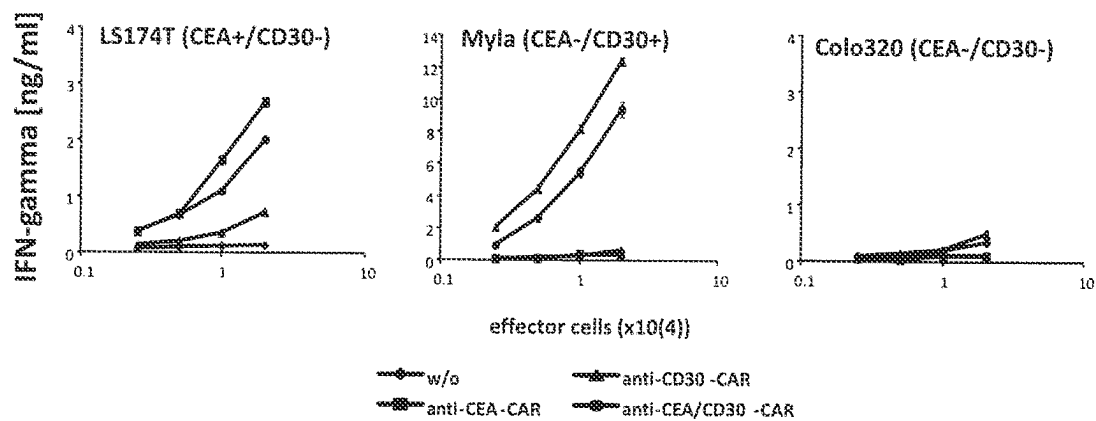


FIGURE 5

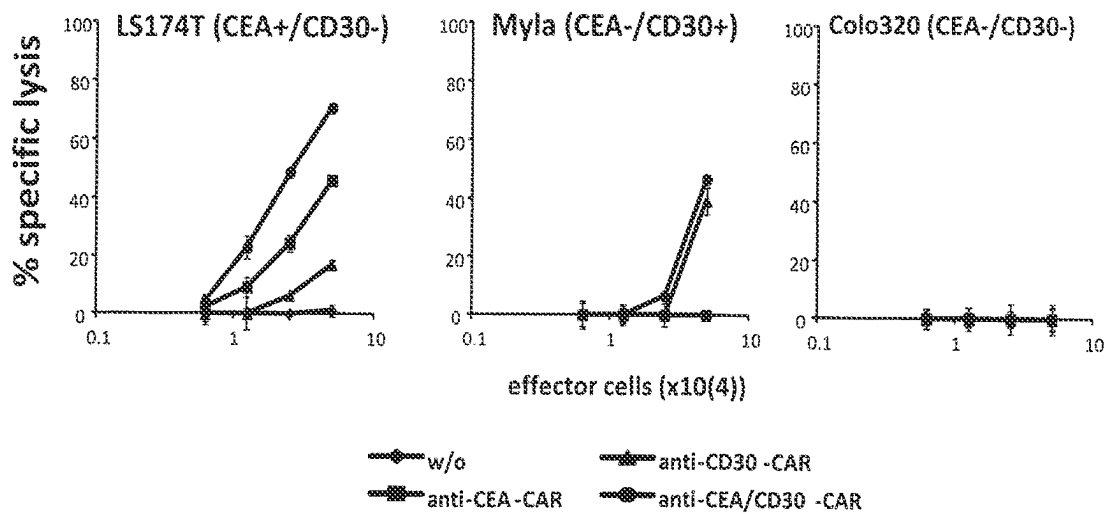


FIGURE 6

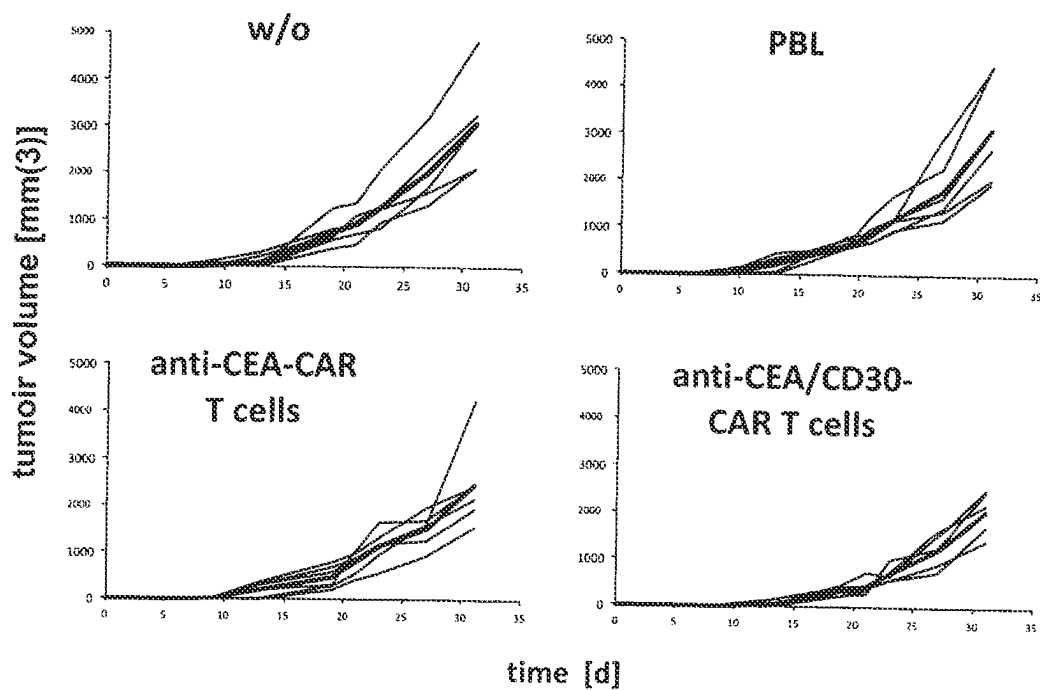


FIGURE 7A

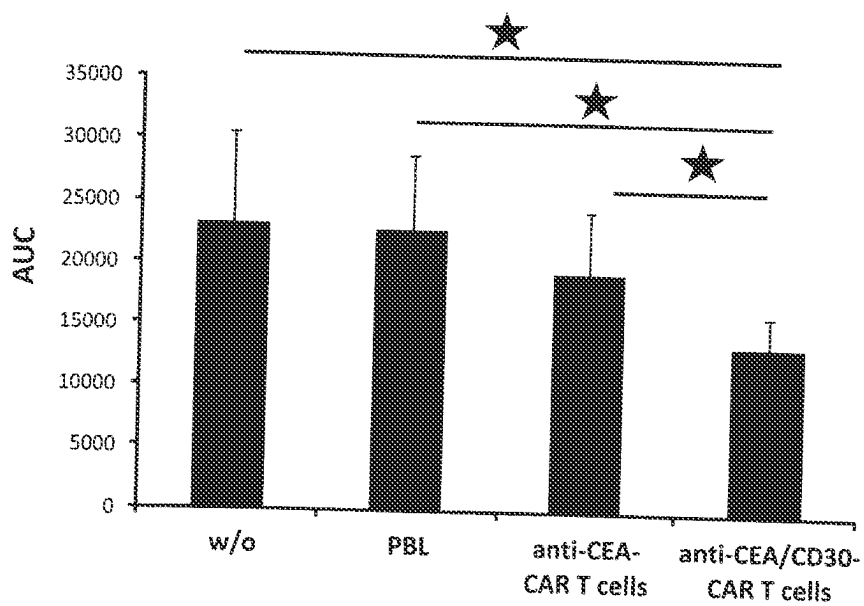


FIGURE 7B

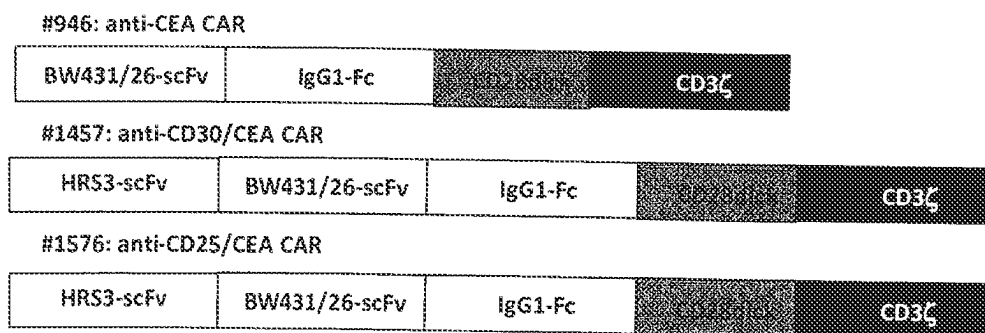


FIGURE 8A

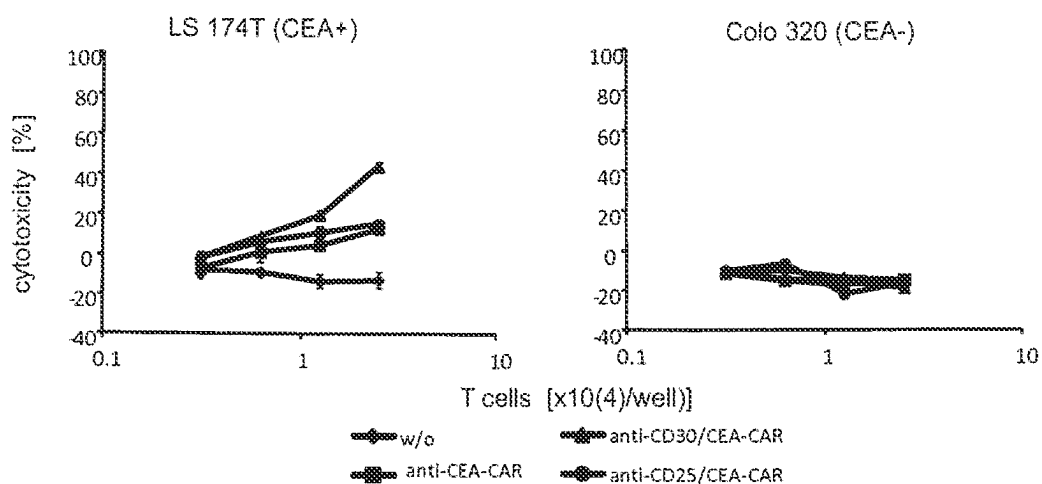
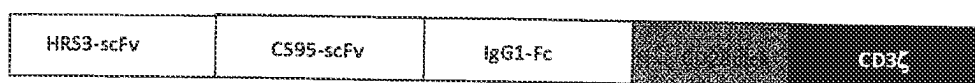


FIGURE 8B

#1587: anti-CD30/Muc1 CAR



#1650: anti-CD30/TAG-72 CAR

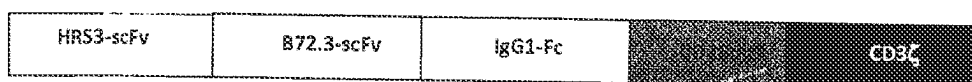


FIGURE 9A

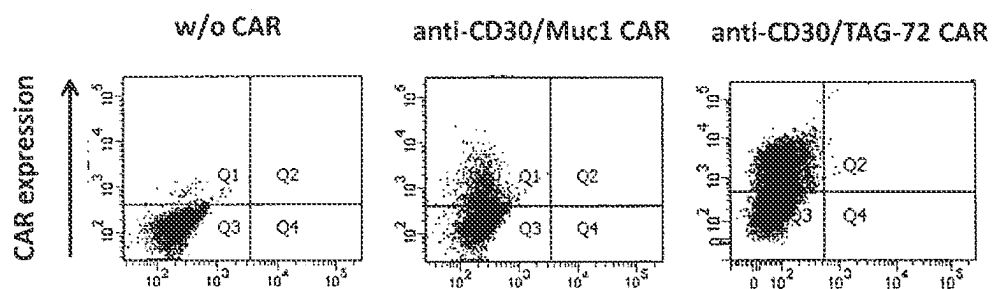


FIGURE 9B

CHIMERIC ANTIGEN RECEPTOR AND ITS USE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation application of U.S. Ser. No. 15/325,638 filed Jan. 11, 2017, now U.S. Pat. No. 10,865,243, which was a 35 U.S.C. 371 national stage filing of International Application No. PCT/EP2015/066252 filed Jul. 16, 2015, which claimed priority to European Application 14177278.0 filed Jul. 16, 2014.

FIELD OF THE INVENTION

In a first aspect, the present invention relates to a recombinant polypeptide containing a domain comprising at least two antibody units whereby one antibody unit is an anti-CD30 single chain antibody unit while another antibody unit is an antibody unit being specific for an antigen present on the surface of a predetermined target cell. In particular, the present invention relates to a recombinant polypeptide containing at least the following domains starting from the N-terminus to the C-terminus: a first domain containing an anti-CD30 single chain antibody unit, in particular, HRS3 scFv of SEQ ID No. 2 or homologous thereof having at least 70% identity with SEQ ID No. 2 binding specifically to CD30, and an antibody unit said antibody unit being specific for an antigen present on the surface of a predetermined target cell, in particular, being specific for a tumor-associated antigen; optionally a spacer domain; a trans-membrane domain; and a cytoplasmatic signalling domain. The recombinant polypeptide can likewise be arranged in the order where the domain comprising at least two antibody units has the order: a first domain containing an antibody unit specific for a tumor-associated antigen and the second domain containing an anti-CD30 single chain antibody unit. In a further aspect, the present invention relates to a nucleic acid molecule encoding the polypeptide according to the present invention, as well as vectors and cells containing the same. Moreover, lymphocytes are provided, in particular T-cells like CD8₊ or a CD4₊ T-cell, expressing on its surface chimeric antigen receptors containing an anti-CD30 single chain antibody unit and an antibody unit whereby said antibody unit being specific for an antigen present on the surface of a predetermined target cell.

These cells are particularly useful in treating cancer, in particular CD30⁺ cancer. That is, the cells are for use in adoptive cell therapy for treating cancer in a subject in need thereof.

PRIOR ART

Adoptive T-cell transfer has shown significant efficacy in the treatment of malignancies and can be curative in patients with various diseases including leukaemia or Epstein Barr virus associated malignancies. Usually, patient derived T-cells are engineered ex vivo to express a recombinant T-cell receptor or, alternatively, a Chimeric Antigen Receptor. Said Chimeric Antigen Receptor (CAR) is typically composed of an extracellular antigen binding domain derived from an antibody and an intracellular T-cell activation domain derived from the T-cell receptor endodomain. In contrast to the physiological TCR, the CAR is composed of one single polypeptide chain that combines antigen binding via the extracellular moiety with a T-cell activation machinery provided by the intracellular signalling moiety. Thus,

due to the antibody derived binding domain, CAR modified T-cells recognize their target, usually, a cell surface antigen, independently of the Major Histocompatibility Complex (MHC) presentation of antigen and are not compromised by tumour cell variants with lowered or deficient antigen processing which represents a commonly observed mechanism of tumour immune escape. CARs are in the focus of research in the recent years. In particular, their capacity to target and lyse pre-defined cells are in the focus of immunotherapy. Recent clinical trials have underscored the potential of adaptive therapy of cancer with CAR-redirectioned T-cells. For example, neuroblastoma patients treated with GD2 Ganglioside-specific CAR T-cells showed some encouraging anti-tumour effects although the T-cells persist only for a brief period. Further studies proved the concept that CAR engineered T-cells can initiate a productive anti-tumour response in patients suffering from various malignancies. The CAR approach differs to other antibody mediated immune therapy strategies, e.g. by using immunotoxins, in so far that engineered cells are used instead of single molecules.

In recent years, efforts have been done in the optimization of the CAR design, see e.g. Bridgeman J. S., et al., *Curr Gene Ther* 2010, 10, 77-90. However, many challenges remain, in particular, the necessity of a more effected anti-tumour response and prolonging T-cell survival allowing long term T-cell persistence of said engineered T-cells in the body. In addition, the co-stimulatory signals required for a successful clinical application remains to be illusive, e.g. see Hornbach A., et al., *Curr Mol Med*. 2013 August; 13(7):1079-88. Hence, there is ongoing work on optimizing CAR for various approaches including adaptive immunotherapy.

Hornbach A., et al., *Gene Therapy*, 2010, 17, 1206-1213 describe the modification of the IgG1 Fc spacer domain in the extracellular moiety of Chimeric Antigen Receptors for avoiding off-target activation and unintended initiation of an innate immune response. Therein, T-cells engineered with modified CAR, either an anti-CD30 CAR or an anti-CEA CAR, have been used. As demonstrated therein, anti-CEA CAR T-cells are not effective against CEA negative cells and anti-CD30 CAR T-cells not against CD30⁺ cells. The modification disclosed in this publication relates to the avoidance of target activation by unintended initiation of the immune response, e.g. due to cross reactivity with a spacer domain in the extracellular moiety of the CAR.

In Kofler et al., 2011, *Mol. Ther.* 19, 760-767 a CAR molecule is described having a CD28 endodomain combined with a CD3 endodomain and an antibody derived scFv ectodomain specific for CEA. It is described therein, that a deletion of the lek binding moiety in the CD28 CAR endodomain improves redirected antitumour activity in presence of T-regulatory cells without impairment of interferon gamma secretion, proliferation and cytolysis. It is speculated, that the CAR with the modified CD28 endodomain expedite the implementation of adoptive T-cell therapy in patients with a variety of cancer types that are heavily infiltrated by regulatory T-cells (Treg cells).

A summary of the present knowledge on chimeric antigen receptor redirectioned T-cells is given in the review article of Chmielewski M., et al, *frontiers in immunology*, 2013, 4, 1-7 as well as in Melenhorst J J, Levine B L., *Cytotherapy*. 2013 September; 15(9):1046-53. In addition, a summary of adoptive therapy of cancer with CAR redirectioned T-cells is provided in Hornbach A., et al., *Curr Mol Med*, 2013, 13(1), 1-10. Therein, the effects of CAR are summarized including co-stimulation activity as well as improvement and prolongation of the redirectioned anti-tumour T-cell response. In

addition, the adverse effects of this kind of adaptive therapy are described including “cytokine storm” and “T-cell repression”.

Recently, Chmielewski et al, Immunol Reviews vol. 257, 2014, p83-90 discusses the possibility of CAR T-cells engineered with an inducible cytokine to modulate the tumor stroma.

Wilkie, S. et al., J. Clin. Immunol., 2012, 32, 1059-1070 describe a dual targeting of ErbB2 and MUC1 in breast cancer using two separate CARs, each capable for full T cell activation. Kloss et al., Nat Biotechnol. 2013 January; 31 (1):71-5, report T cells engineered with two separate CARs which provide complementary signalling. The vectors described therein encode two separate CAR molecules expressed simultaneously by the same T cells.

Grada, Z. et al., Molecular Therapy-Nucleic Acids, 2013, vol. 2, page 105 relate to TanCAR, a novel bispecific CAR for cancer immunotherapy. The CAR molecules described therein are designed to identify ligands both being present on the target cell.

That is, currently used chimeric antigen receptor (CAR) modified T-cells for antigen-redirected targeting towards tumor cells show insufficient performance in the anti-tumor attack due to various reasons, in particular due to less amplification and cytolytic activity after adoptive transfer into the patient. Hence, a major obstacle of the strategy for clinical use is the insufficient performance of engineered T-cells in the long-term.

Further, beside beneficial effect of the CAR expressing T-cell in adoptive therapy, ample effects are known which presently hinder favourite development of respective therapy as mentioned above.

BRIEF DESCRIPTION OF THE PRESENT INVENTION

The present invention aims at improving persistence and performance of CAR T-cells.

In a first aspect, the present invention relates to a recombinant polypeptide containing at least the following domains starting from the N-terminus to the C-terminus: a first domain containing an anti-CD30 single chain antibody unit, in particular, HRS3 scFv of SEQ ID No. 2 or homologs thereof having at least 70% identity with SEQ ID No. 2 binding specifically to CD30, and an antibody unit said antibody unit being specific for an antigen present on the surface of a pre-determined target cell, in particular, being specific for a tumor-associated antigen; optionally a spacer domain; a trans-membrane domain; and a cytoplasmatic signalling domain. In particular, the present invention relates to a recombinant peptide wherein the first domain comprises the anti-CD30 single chain antibody unit in combination with an antibody unit whereby this antibody unit is specific for a tumor-associated antigen. In a favourable embodiment, the anti-CD30 single chain antibody unit is the HRS3 scFv of SEQ ID No. 2 while the antibody unit is a unit binding to the tumor-associated antigen of carcinoembryonic antigen (CEA), in particular of SEQ ID No. 4. In a further aspect, the present invention relates to a nucleic acid molecule encoding the polypeptide according to the present invention as well as a vector comprising said nucleic acid sequence. Moreover, a cell, cell line or host cell containing said nucleic acid sequence or said vector is provided.

In addition, the present invention is directed to a lymphocyte, in particular, a T-cell, like a CD8₊ and/or a CD4₊ T-cell, expressing on its surface chimeric antigen receptors having in their extracellular domain two antibody units, an

antibody unit being specific for an antigen present on the surface of a predetermined target cell, and an anti-CD30 single chain antibody unit. These units are either present in a single polypeptide as defined herein or present in different functional chimeric antigen receptors. These cells are particularly useful for treating tumor cells expressing the antigen on its surface but may not express CD30 on its surface. That is, by combining these two units, namely, the anti-CD30 single chain antibody unit and an antibody domain being specific for a tumor-associated antigen, tumor cells expressing the tumor specific antigen on its surface but being CD30⁺ can be treated more efficiently. Hence, the present invention provides in a further aspect a method for treating subjects being afflicted with cancer whereby the tumor cells of said cancer express the antigen but are CD30⁺(CD30 negative).

In particular, the cells useful for adoptive cell therapy according to present invention express on its surface chimeric antigen receptors having in the extracellular domain the antibody unit being specific for an antigen present on the surface of a predetermined cell, in particular, wherein the antibody unit is an antibody unit being specific for a tumor-associated antigen, and an anti-CD30 single chain antibody unit whereby these two units may be present either in a single polypeptide representing a CAR or may be present in different functional chimeric antigen receptors.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1: Schematic representation of the modular composition of the CAR molecules. #1138 represents the anti-CD30 CAR; #946 represents the anti-CEA CAR. The CAR #1457 is the bispecific CAR molecule according to the present invention composed of the anti-CD30 single chain antibody scFv HRS3, the anti-CEA scFv BW431/26, the spacer domain IgG1-Fc, the trans-membrane domain of CD28 and the cytoplasmic signalling domain of the CD28ALCK and CD3zeta chain.

FIG. 2: Peripheral blood lymphocytes were engineered with CARs by standard procedures. After cultivation CD3⁺ cells were analysed for CAR expression utilising FITC-conjugated anti-CD3 and PE-conjugated anti-human IgG1Fc antibodies. As demonstrated, transduction was successfully conducted and T-cells express the anti-CD30/CEA CAR on the cell surface in similar amounts as T-cells transduced with the anti-CD30 and anti-CEA CAR, respectively.

FIG. 3: Human T-cells were engineered with CARs as indicated with an initial CAR expression rate of anti-CEA-CAR: 22.2%; anti-CD30-CAR: 28.9%; anti-CEA/CD30-CAR: 22.8%. CAR T-cells were cultivated 300 hours in the presence of IL-2 and CAR expressing cells were monitored. CAR T-cells with a CD30 binding domain expand more efficiently than T-cell with the anti-CEA CAR only.

FIG. 4: Engineered human T-cells expressing different kinds of CARs as indicated (initial CAR expression: anti-CEA-CAR: 8.5%; anti-CD30-CAR: 9.3 9%; anti-CEA/CD30-CAR: 8.2%). CAR T-cells (0.625×10^4 cells/well) were co-incubated for 24 hours with tumor cells (2.5×10^4 cells/well) as indicated. Three different types of cell lines were used being either positive for CEA or CD30 or as control negative for both CEA and CD30. The cell line being negative for both CEA and CD30 was not lysed while specific lysis of LS174T cells was observed which is a CEA7CD30⁺ cell line. Therein, the anti-CEA/CD30 CAR T-cells demonstrate a higher specific lysis compared to the anti-CEA CAR T-cells.

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FIG. 5: The release of interferon-gamma, a marker for T-cell activation, during co-cultivation of different numbers of effector cells with the tumor cells is shown. After 24 hours co-cultivation the supernatants were collected and analysed for interferon-gamma by ELISA. The assay was performed in triplicate and the SD was determined. As shown, anti-CEA/CD30 CAR T-cells produced IFN-gamma when co-incubated with the CEA⁺CD30⁺LS174T-cells as well as with the CEA⁺CD30⁺ cells. No IFN-gamma was produced upon co-incubation with CEA⁺CD30⁺ Colo320 cells. T-cells with the anti-CEA CAR or anti-CD30 CAR released IFN-gamma only upon co-incubation with CEA⁺ or CD30⁺ cells, respectively.

FIG. 6: FIG. 6 shows the cytolytic activity of CAR CIK T-cells (cytokine induced killer cells with NKT-cell phenotype). It is demonstrated that enhanced killing of CEA⁺CD30⁺ target cells by using the anti-CEA/CD30 CAR CIK T-cells is achieved. The tumor cell lines are the same as described in FIG. 2.

FIGS. 7A-7B: FIG. 7A demonstrates improved in vivo killing of CEA⁺ tumor cells by anti-CEA/CD30 CAR T-cells compared to the mono-specific anti-CEA CAR T-cells. Sub-therapeutic doses of T-cells (2.5×10^5 cells/mouse) without CAR and T-cells engineered either with the anti-CEA/CD30 CAR or the anti-CEA CAR were co-injected with CEA⁺CD30⁺ 015A3 tumor cells (1×10^6 cells/mouse) into immune deficient Rag common gamma chain mice. For comparison, CI 5A3 tumor cells were injected without T-cells (w/o). Tumor volumes and area under curve (AUC) were determined (FIG. 7B). Significance was determined by Student's T-test. Significant differences ($p < 0.05$) were indicated by asterisks.

FIGS. 8A-8B: FIG. 8A shows schematic representation of anti-CD30 CARs and the CD25/CEA bispecific CAR of Seq. ID. No. 18 encoded by the sequence of Seq. ID. No. 17. The HRS3-scFv of CAR #1457 was exchanged by the anti-CD25-scFv RFT5. The backbone of the resulting #1576 CAR is identical to CAR #1457 shown in FIG. 1. FIG. 8B shows the specificity of cytotoxicity. CART cells (0.625×10^4 cells/well) were co-incubated for 24 hours with CEA⁺ or CEA⁻ tumor cells (2.5×10^4 cells/well) as indicated. Only CAR T cells expressing the bispecific anti-CD30/CEA CAR lysed CEA⁺ target cells whereas CAR T cells with the bispecific anti-CD25/CEA CAR and CAR T cells with a monospecific anti CEA CAR lysed CEA⁺ targets with lower efficiency. CEA-tumor cells were not lysed demonstrating the CEA-specificity of CAR T cells.

FIGS. 9A-9B: FIG. 9A is a schematic representation of the molecular composition of the CAR molecules with specificity for CD30/MUC1 of Seq. ID. No. 20 encoded by the sequence of Seq. ID. No. 19 and CD30/TAG-72 of Seq. ID. No. 22 encoded by the sequence of Seq. ID. No. 21. B72.3 is used for the anti-TAG-72 CAR. Except for the anti-CEA scFv, the molecules are identical with the CAR shown in FIG. 1. Cells surface expression of CD30/Muc1 and CD30/TAG-72 CARs is demonstrated in FIG. 9B by flow cytometry.

DETAILED DESCRIPTION OF THE PRESENT INVENTION

The inventors aim to provide new recombinant polypeptides containing at least the following domains starting from the N-terminus to the C-terminus: a first domain containing an anti-CD30 single chain antibody unit, in particular, HRS3 scFv of SEQ ID No. 2 or homologous thereof having at least 70% identity with SEQ ID No. 2 binding specifically to

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CD30, and an antibody unit said antibody unit being specific for an antigen present on the surface of a predetermined target cell, in particular, being specific for a tumor-associated antigen; optionally a spacer domain; a trans-membrane domain; and a cytoplasmatic signalling domain. These recombinant polypeptides represent bispecific CAR peptides. In an embodiment of the present invention, these CAR polypeptides are expressed by genetically engineered T-cells containing and eventually expressing these CAR polypeptides.

That is, in a further aspect, the present invention relates to a recombinant polypeptide being at least the following domains starting from the N-terminus to the C-terminus: a first domain containing an anti-CD30 single chain antibody unit, in particular, HRS3 scFv of SEQ ID No. 2 or homologous thereof having at least 70% identity with SEQ ID No. 2 binding specifically to CD30, and an antibody unit said antibody unit being specific for an antigen present on the surface of a predetermined target cell, in particular, being specific for a tumor-associated antigen; optionally a spacer domain; a trans-membrane domain; and a cytoplasmatic signalling domain.

The present invention likewise relates to the following domains, a first domain an antibody unit said antibody unit being specific for an antigen present on the surface of a predetermined target cell, and a second domain containing an anti-CD30 single chain antibody unit, optionally a spacer domain; a trans-membrane domain; and a cytoplasmatic signalling domain.

As used herein, the term "comprise" or "comprising" as well as the terms "contain" or "containing" includes the embodiment of "consist" or "consisting".

The term "homolog" as used herein refers to molecules, either DNA or polypeptides, having a sequence homology of a certain amount, namely of at least 70%, like at least 80%, 90%, 95%, 96%, 97%, 98%, 99% of the nucleic acid sequence or the amino acid sequence it is referred to. Homology refers to the magnitude of identity between two sequences. Homolog sequences have the same or similar characteristics, in particular, have the same or similar property of the sequence as identified. For example, the homolog of the HRS3 scFv sequence of Seq. ID. No. 2 has the same or similar binding specificity to the CD30 molecule as it is the case for the HRS3 scFv molecule. Further, homologs include nucleic acid molecules encoding the same peptide but may vary in its sequence due to the degeneracy of the genetic code. Further, "identify" refers to the presence of identical amino acid or nucleic acid molecules in the order as described for the sequence it refers to. That is, in case of at least 90% identity, 90% or more of the nucleic acid and amino acid molecules, respectively, are identical at the respective positions. Unless otherwise identified, the terms "homology" and "identity" are used herein interchangeably. In particular, the homolog of the HRS3 scFv sequence of Seq. ID. No. 2 include anti CD30 single chain antibody units binding to the same epitope recognized by the HRS3 scFv.

In addition, the term "genetically engineered" refers to cells being manipulated by genetic engineering. That is, the cells contain a heterologous sequence which does not naturally occur in said cells. Typically, the heterologous sequence is introduced via a vector system or other means for introducing nucleic acid molecules into cells including liposomes. The heterologous nucleic acid molecule may be integrated into the genome of said cells or may be present extra-chromosomally, e.g. in the form of plasmids. The term also includes embodiments of introducing genetically engineered, isolated CAR polypeptides into the cell.

Generally, CARs are fusion proteins, consisting of an extracellular antibody type recognition domain fused to intracellular T-cell signalling proteins. Typically, the ectodomain containing the antigen recognition region comprises a signal peptide and an antigen recognition unit. According to the present invention, the ectodomain comprises an anti CD30 single chain unit in combination with an antibody unit being specific for an antigen present on the surface of a predetermined target cell, in particular, being specific for a tumor-associated antigen. It is preferred, that said single chain unit is a single chain unit selected from HRS3 scFv of SEQ. ID. No. 2 or homologous thereof having at least 70% identity with SEQ ID No. 2 and binding specifically to CD30. Further, the single chain unit may be derived from other anti-CD30 antibodies like HRS4 or Ki-4. Said antibodies have the same binding specificity to the same epitope of CD30 as it is the case for the HRS3 antibody. That is, in an embodiment of the present invention, it is preferred that the anti-CD30 single chain unit is a single chain unit being specific to the epitope of the CD30 molecule recognized by the anti CD30 single chain unit HRS3 scFv of Seq. ID. No. 2.

The ectodomain may be spaced apart from the transmembrane domain by the presence of a spacer domain. Said optional spacer domain links the antigen-membrane domain is flexible enough to allow the antigen binding domain to orient in different directions to facilitate antigen recognition.

The transmembrane domain is typically a hydrophobic alpha helix that spans the membrane. Finally, the endodomain represents the signalling domain in the cytoplasm of the cells.

Further, the term "antibody unit being specific for an antigen present on the surface of a predetermined target cell" refers to the binding site derived from an antibody of any species either in a binding domain to the transmembrane domain and it is preferred that said transform of a single polypeptide chain or multiple covalently or non-covalently associated polypeptide chains. These domains are characterized by specific binding to a molecule on the surface of a target cell. Any other binding moiety than an antibody can alternatively be used as targeting domain in a CAR, e.g., the binding moiety of a receptor like a cytokine receptor.

Moreover, the term "antibody unit being specific for a tumor-associated antigen" refers to a binding domain with specificity to a target molecule present on the cell surface expressed by tumor cells and composed of a polypeptide, carbohydrate or lipid or combinations thereof that are exclusively or preferentially expressed by malignant cells.

The term "functional CAR" refers to a CAR expressed by immune cells that by specific binding to the cognate target molecule activates the same immune cell to increase protein biosynthesis, cytokine secretion, cell proliferation and target cell lysis.

Moreover, the term "CD30₊ cells" or "CD30₊ cancer cells" refers to cells (either non-cancer or cancer cells) expressing on their surface the CD30 molecule.

The terms "non-tumor cells" and "tumor cells" as well as "non-cancer cells" and "cancer cells" are used herein interchangeably unless otherwise defined.

The term "CIK T-cells" or "NK T-cells" which are used herein interchangeably refers to a heterogeneous group of T-cells that share properties of both T-cells and natural killer (NK) cells. CAR CIK T-cells are obtained by known cultivation techniques to obtain firstly NK T-cells or CIK T-cells and, thereafter, engineering the same cells with the CAR by known methods. The recombinant polypeptide according to the present invention represents a bispecific CAR also

identified herein as dual specific CAR. In an embodiment, the order in the first domain is from the N-terminus to the C-terminus: i) an anti-CD30 single chain antibody unit and an antibody unit, said antibody unit being specific for an antigen present on the surface of a predetermined target cell, or ii) an antibody unit whereby said antibody unit being specific for an antigen present on the surface of a predetermined target cell, and an anti-CD30 single chain antibody unit.

It is preferred that the antibody unit being specific for an antigen present on the surface of a predetermined target cell is an antibody unit being specific for a tumor-associated antigen.

In an embodiment of the present invention, the CAR present in the T-cell comprises a leader sequence being located N-terminally to the first domain containing the anti CD30 single chain unit and the antibody unit.

In addition, in another embodiment, the anti-CD30 single chain antibody unit is a HRS3 scFv peptide, in particular, of SEQ. ID. No. 2, e.g. encoded by the nucleic acid sequence of Seq. ID. No. 1. It has been recognized herein that an anti-CD30 single chain antibody fragment of the variable region (scFv), in particular, of HRS3, allows displaying the desired activity. In an embodiment, the anti CD30 single chain antibody fragment is a fragment recognizing and binding to the epitope of HRS scFv.

In another embodiment, the spacer domain of the CAR molecule is an IgG hinge-CH2CH3 domain of SEQ. ID. No. 6 or homologs thereof having at least 70% identity therewith, preferably, the spacer domain is a mutated IgG hinge-CH2CH3 domain according to SEQ. ID. No. 6, e.g. encoded by the nucleic acid sequence of Seq. ID. No. 5.

In some embodiments, a linker may be located between the spacer domain and the transmembrane domain.

Further, another embodiment relates to a T-cell with a chimeric antigen receptor wherein the transmembrane domain is derived from the CD28 molecule, e.g. the transmembrane domain of the CD28 molecule lacking the ICD domain of SEQ. ID. No. 8, e.g. encoded by the nucleic acid sequence of Seq. ID. No. 7.

The signalling domain or endodomain or intracellular domain which are used herein interchangeably, contains a CD3 zeta or FcEpsilon-Receptor gamma chain signalling chain or a co-stimulatory domain or both the CD3zeta and a costimulatory domain or the FcEpsilon-Receptor gamma chain and a co-stimulatory domain. For example, the intracellular domain is a CD3 zeta signalling domain of SEQ. ID. No. 10, e.g. encoded by the nucleic acid sequence of Seq. ID. No. 9, or a homolog thereof having at least 70% homology. In another embodiment, the intracellular domain is the IgE Fc-Receptor gamma signalling domain of SEQ. ID. No. 12, e.g. encoded by the nucleic acid sequence of Seq. ID. No. 11, or a homolog thereof having at least 70% identity. The signalling domain is responsible for the activation of the T-cells, in particular the cytotoxic activity and cytokine release including Interferon gamma secretion.

The CAR molecule may be a so-called second generation CAR molecule. Second generation CAR molecules have improved signalling domains additionally containing a second signalling (costimulatory) domain, e.g. derived from CD28, CD134 (OX40) or CD137 (4-1BB). Third generation CAR molecules contain a combined co-stimulatory signalling domain, e.g., CD28 combined with CD137 or CD134.

An overview about the CAR molecules is provided e.g. in Gilham D. E. et al., Trans, and Molecular Medicine, 2012, 18(7), 377-384.

In a preferred embodiment of the present invention, the T-cell is a T-cell with a chimeric antigen receptor wherein the chimeric antigen receptor is a polypeptide of SEQ. ID. No. 14, e.g. encoded by the nucleic acid sequence of Seq. ID. No. 13. Said CAR is also referred to herein as #1457.

The anti-CD30 anti-CEA CAR #1457 is expressed on the surface of T-cells and is composed in the extracellular part of the anti-CD30 single chain fragment of variable region (scFv) antibody HRS3, the BW431/26 scFv antibody corresponding to Seq. ID. No. 4, e.g. encoded by the nucleic acid sequence of Seq. ID. No. 3, corresponding to the anti-CEA antibody BW431/26 and the modified human IgG1 CH2CH3 domain as spacer between scFv and the trans-membrane domain. The modification of the IgG1 domain consists of point mutations to convert the wild-type amino acid sequence PELLGGP X13 MISRT (Seq. ID. No. 15) to PPVA-GP X13 MIART (Seq. ID. No. 16) which reduces unintended binding of the CAR Fc domain to Fc receptors on other cells like innate immune cells which would mediate their activation. The transmembrane and intracellular membrane proximal part of CAR #1457 is derived from human CD28 and is fused to the intracellular part of human CD3zeta. The CD28 sequence is mutated at P560>A560, P563>A563, P564>A564 (Kofler et al., Mol. Ther. 19, 760-767 (2011)). Thereby the CD28 binding site for the lek kinase is destroyed with the consequence that activation of the lek signalling pathway and subsequent CAR mediated IL-2 secretion is prevented. Pre-clinical models imply that Treg cell mediated repression of CAR T- cell effector functions is reduced under these conditions.

As an example, the recombinant polypeptide is the polypeptide of SEQ ID No. 14, e.g. encoded by the nucleic acid sequence of Seq. ID. No. 13 correspond to the #1457 CAR shown in FIG. 1.

In another embodiment, the recombinant polypeptide is a polypeptide wherein the antibody unit binds to a tumor-associated antigen including any one of carcinoembryonic antigen (CEA), CA19-9, CA72-4 also known as TAG-72, PSCA, Muc-1, HMW-MAA, p97 melanotransferrin, fetal actelycholin receptor, ErbB2 (Her2/neu), multi-drug-resistance protein (MDR), CD19, CD20, TOSO.

For example, the CAR #1457 of Seq. ID. No. 14 correspond to a polypeptide wherein the antibody unit being specific to CEA. Further, CARs useful according to the present invention are shown in FIG. 9, namely, CD30/MUC-1 CAR or CD30/Tag72 CAR. These CARs are #1587, anti-CD30/MUC1, of Seq. ID. No. 20 encoded by the sequence of Seq. ID. No. 19, and #1650 anti-CD30/TAG-72 of Seq. ID. No. 22 encoded by the sequence of Seq. ID. No. 21.

In another aspect, the antibody unit binds to viral antigens, in particular, hepatitis virus B-associated antigen S or L, cytomegalovirus-associated antigen or other viral antigens described in the art as being useful for detecting and, eventually, destroying virus infected cells.

As demonstrated in the examples, T-cells (either CD4 or CD8 T-cells) or CIK-cells demonstrate specific lysis of cells not expressing CD30 but expressing the antigen to which the antibody unit is specifically binding only. That is, surprisingly the percentage of specific lysis is higher with the bispecific CAR molecule compared to the monospecific molecule containing the antibody binding unit specific for the antigen present on the surface of the predetermined target cell only.

It is preferred, that the target cell is a tumor cell. Further, the target cell may be a virus loaded cell, e.g. a cell infected with a virus.

Moreover, the predetermined target cell may be a cell involved in autoimmune defects in a subject. In that particular case suppressor cells, in particular regulatory T-cells, are modified with the described CAR in order to repress the acute inflammatory immune reaction in the targeted tissue. The CAR binds to tissue antigen with one scFv, e.g., anti-HLA B27, and to CD30 with the other scFv.

Although not expressed by the predetermined target cells, the anti-CD30 single chain antibody unit present in the CAR molecule increases specific lysis of the target cells. Furthermore, it has been surprisingly shown that in case of CD30 negative cells the percentage of specific lysis can be increased as demonstrated in the examples. This is not only true for lysis by CAR expressing T-cells but also by CAR CIK-cells. It is considered that autostimulation of the CAR T-cells allows to increase specific lysis of the target cells.

In addition, the present invention provides nucleic acid molecules comprising the nucleic acid sequence encoding the polypeptide according to the present invention. Furthermore, vectors are provided comprising the nucleic acid sequence according to the present invention encoding the polypeptide as described. The skilled person is well aware of suitable vector systems and vectors, in particular, vectors allowing transfection and transduction of eukaryotic cells, in particular, T-cells.

Moreover, the present invention provides a cell, cell line or a host cell containing the vector according to the present invention or a nucleic acid molecule according to the present invention. Preferably, said cell, cell line or host cell is a T-cell, e.g., a CDC T-cell or a CD8₊ T-cell.

Further, the present invention provides a kit or system containing the vector according to the present invention, the cell, cell line or host cell according to the present invention, or the polypeptide according to the present invention or a nucleic acid molecule according to the present invention or mixtures thereof for use in the production of T-cells expressing the chimeric antigen receptor. The kit or system according to the present invention may contain further components including means for introducing the vector, polypeptide or nucleic acid molecules into the cells. The skilled person is well aware of suitable means for doing so.

Moreover, the present invention provides lymphocytes, in particular, T-cells, like CD8 and/or CD4 T-cells expressing on its surface chimeric antigen receptors having in the extracellular domain an antibody unit, said antibody unit being specific for an antigen present on the surface for an antigen present on the surface of a predetermined target cell, and an anti-CD30 single chain antibody unit expressed by said cells. The units may be present either in a single polypeptide, e.g. a polypeptide as defined herein, or, alternatively, may be present in different chimeric antigen receptors.

In another aspect, the lymphocyte may be a CAR CIK-T-cell as defined herein.

The lymphocytes including the CD8₊, CD4₊ or CIK-T-cells demonstrate higher specific lysis of the target cells, thus, reducing side effects and, in addition, overcoming the problem of insufficient performance in the anti-tumor attach due to various reasons, for example, less amplification and cytolytic activity after adopted transfer into the patient.

It is considered that the presence of the anti-CD30 single chain antibody unit allows to amplify the genetically engineered lymphocytes and improve persistence of the same, thus, overcoming the problem of insufficient performance.

The cells are particularly useful in treating cancer, in particular CD30⁺ cancer whereby said cancer is a cancer

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expressing on the cancer cells the antigen to which the anti-tumor antigen antibody is directed to.

In a further aspect, the present invention relates to the use of immune cells, e.g. the T-cell with a chimeric antigen receptor according to the present invention in adaptive cell therapy for treating CD30⁺ or CD30⁻ cancer in a subject in need thereof. For instance the CD30⁺ cancer may be Hodgkin's lymphoma, anaplastic large cell lymphoma, acute lymphocytic leukaemia, cutaneous lymphoma, mycosis fungoides, lymphoproliferative diseases, systemic mastocytosis, teratocarcinoma, stem cell derived malignancies, or cancer stem cells or others. In particular the invention relates to the use of such immune cells for the treatment of CD30⁻ cancers, e.g., breast, lung, prostate, pancreas, gastrointestinal carcinomas, neuronal cancer and others. Thus, the present invention may foster adaptive immune therapy by CAR T-cells.

The present invention is further described by way of examples. Said examples illustrate the invention further without limiting the same thereto.

EXAMPLES

Preparation of CAR #1457

The expression cassette of the bispecific anti-CD30/CEA-CAR was generated as follows: The anti-CD30-scFv (HRS3) and the anti-CEA-scFv (BW431/26) were PCR amplified and flanked with a linker by overlapping sequences at the 3'terminus of the CD30-scFv and the 5'terminus of the anti-CEA-scFv while the linker coding for a 40 AA (Gly(4)-Ser(1))₅ linker. DNAs of scFvs were assembled by SOE PCR, digested with NcoI and BamHI and ligated into the NcoI/BamHI site of the retroviral vector pBullet-HRS3-scFv-Fc-CD28deltaIck-zeta thereby replacing the HRS3-scFv by the combined HRS3scFv-BW431/26scFv. The resulting vector coding for the bispecific anti-CD30/CEA-CAR was termed #1457. The modular composed #1457 CAR and its mono-specific anti-CEA (#946) and anti-CD30 (#1138) counterparts are schematically shown in FIG. 1.

The retroviral vector coding for the #1457 CAR was produced according to SOP-GL-VectProd using a Galv pseudotyped envelope. In summary vector particle production was done transiently on the human embryonic kidney cell line 293T after Polyfect® mediated DNA transfection. Vector particles were pseudotyped with Galv. No vector titer was determined.

Transduction of human blood lymphocytes was done according to SOP standard techniques. In summary human lymphocytes were transduced with a 2-day supernatant from transfected 293T-cells for 2 days. The CAR #1457 was expressed by 20-35% of human T-cells as measured at day 2 by flow cytometry using an antibody directed to the extracellular constant IgG1 CH2 CH3 domain of the CAR.

CD4⁺ and CD8⁺ T-cells which express the CAR #1457 on the cell surface can be recorded by use of the 9G10 antibody which binds specifically the CAR HRS3 scFv domain. T-cells engineered with the #1457 CAR bind specifically to CD30 expressing cells and become activated indicated by increased secretion of cytokines including IFN- γ , by increase in proliferation and in cytolysis of CD30⁺ target cells. Noteworthy, only background levels of IL-2 are secreted when T-cells are stimulated by the CAR. IL-2, however, is secreted in physiological amounts when T-cells are stimulated by their physiological TCR and CD28. Activation of the T-cells #1457 is antigen-specific as defined by the specificity of the CAR since CD30⁻ cells do not trigger

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T cell activation. Soluble CD30, which accumulates in the serum of CD30⁺ lymphoma patients, does not block CAR mediated T-cell activation in concentrations up to 10 pg/ml [Hornbach A, et al., Cancer Res. 1998 Mar. 15; 58(6):1116-9]. This is due to the fact that the CAR must be cross-linked by binding the multiple copies of the targeted antigen in order to trigger T-cell activation which can only occur when CD30 is immobilized or expressed on the surface of target cells but does not occur when CD30 protein is present in solution.

Example 1

Activity of Dual Specific CAR Modified T-Cells
Toward CEA⁺ and/or CD30⁺ Cell Lines

Engineering of T-cells with CAR #1457 was performed as described above. The CAR was detected on the T-cell surface by flow cytometry utilizing an antibody against the Fc domain in the extracellular moiety of the CAR (FIG. 1,2). The anti-CEA/CD30 CAR was expressed with similar efficiency as the mono-specific CARs with either the CD30 or the CEA binding domain. To test for dual specificity of the #1457 CAR, grafted T-cells were co-incubated either with CD30⁺ or CEA⁺ target cells and redirected cytotoxicity and IFN-gamma secretion were monitored. Whereas T-cells with anti-CEA and anti-CD30 CAR were only activated by CEA⁺ or CD30⁺ target cells, respectively, the bispecific CAR activates T-cells after coculture with both CEA⁺ CD30⁺ and CEA⁻ CD30⁺ tumor cells. T-cell activation results in specific target cell lysis and IFN-gamma secretion (cf. FIG. 4, 5 and Example 3).

Conclusion

T-cells with the #1457 CAR harbour dual specificity for binding, i.e., for CD30 and CEA. T-cells expressing the bispecific CAR were activated by engaging target cells with one cognate CAR antigen only as well as by target cells with both anti gens.

Example 2

Specific Expansion of #1457 CAR Modified
T-Cells

T-cells were engineered with the #1457 CAR or for comparison with the corresponding single CD30 or CEA specific CAR, respectively, and cultivated in presence of IL-2 (400 U/ml). The initial number of CAR⁺ T-cells was 20-30%. Cells were tested at different time points for CAR expression by flow cytometry. As summarized in FIG. 3, the number of T-cells with the anti-CD30-CAR and the anti-CEA/CD30 CAR #1457 increased up to 90% of T-cells during cultivation whereas the number of anti-CEA CAR T-cells did not exceed 50%.

Conclusion

Data demonstrate that CAR T-cells with the CD30-binding domain expand superior compared to CAR T-cells with a CEA binding domain. The superior effect is also present when the anti-CD30 binding domain is linked with the anti-CEA binding domain.

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Example 3

Increased Specific Lysis by CAR #1457 Engineered
T-Cells and CAR #1457 Engineered CIK T-Cells
Toward CD30-Negative Target Cells

(a) T-Cells

Human T-cells were engineered to express the CAR #1457, the CEA-specific CAR and the CD30-specific CAR, respectively. The same number of CAR T-cells was co-incubated with tumor cells which express either CEA or CD30 or both. Data presented in FIG. 4 show that the cell line being negative for both CEA and CD30 (Colo320) was not lysed while specific lysis of LS174T cells was observed which is a CEA₊/CD30⁺ cell line. Therein, the anti-CEA/CD30 CAR T-cells demonstrate a higher specific lysis compared to the anti-CEA CAR T-cells.

The cytolytic activity is accompanied by release of IFN- γ , a marker for T-cell activation, as shown in FIG. 5. Anti-CEA/CD30 CAR #1457 T-cells produced IFN- γ when co-incubated with the CEA₊/CD30⁺ LS174T-cells as well as with the CEA⁺ CD30⁺ cells. No IFN- γ was produced upon co-incubation with CEA⁺ CD30⁺ Colo320 cells. T-cells with the anti-CEA CAR or anti-CD30 CAR released IFN- γ only upon co-incubation with CEA₊ or CD30₊ cells, respectively.

(b) CIK Cells

CIK cells were generated according to standard procedures and were engineered with the anti-CD30/CEA CAR #1457. After 8 days post transduction #1457 CAR CIK cells were co-cultivated with CD30 CEA₊, CD30 CEA⁺ or CD30 CEA⁺ target cells and target cell lysis was recorded. CIK cells expressing the corresponding monospecific anti-CD30 and anti-CEA CAR, respectively, served as control. Bispecific anti-CD30/CEA and mono-specific anti-CD30 CAR T-cells lysed CD30₊ target cells with similar efficiency. In contrast, CAR #1457 CIK cells lysed CEA₊ CD30⁺ target cells more efficiently than CIK cells with the anti-CEA CAR (FIG. 6). Data indicate that CAR #1457 CIK cells surprisingly show a higher cytolytic activity against tumor cells that do not express CD30 than CIK cells with a CEA-specific CAR. Conclusion:

T-cells and CIK cells engineered with the CAR #1457 show enhanced anti-tumor reactivity against target cells, both which express and which lack CD30, compared to T-cells with the monospecific CAR. This is unexpected in particular for those target cells which lack CD30.

Example 4

Improved Anti-Tumor Activity of #1457 CAR
T-Cells in a Xenograft Mouse Model

The in vivo activity of anti-CD30/CEA CAR #1457 T-cells was monitored in the immune deficient Rag common gamma chain $\gamma^{-/-}$ mouse. Anti-CEA-CAR and anti-CD30/CEA CAR #1457 T-cells were engineered as described above and subcutaneously co-injected in a sub-therapeutic dose (2.5×10^6 CAR T-cells/animal) with CEA₊ CD30⁺ C15A3 tumor cells (1×10^6 cells/animal), that were transfected to express human CEA, into mice. Mice without T-cells and non-modified T-cells were used for control and tumor growth was monitored every 2nd-3rd day. Growth curves (FIG. 7A) and area under curve (FIG. 7B) were determined. Whereas anti-CEA-CAR T-cells produced slightly delay of tumor growth, anti-CD30/CEA CAR T-cells significantly delayed tumor growth indicating higher

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anti-CEA activity of the bispecific CAR T-cells #1457 in vivo than T cells with the anti-CEA CAR.

Conclusion

Bispecific anti-CD30/CEA CAR T-cells are more effective in an prolonged anti-tumor response against CEA₊ CD30⁺ tumor cells in vivo compared to T cells with the mono-specific anti-CEA CAR. The improved anti-tumor reactivity is due to the particular CAR design because human CD30 is neither expressed by the murine host nor by the grafted mouse tumor cells and not recognized on tumor cells or on cells of the host.

Example 5

Improved Anti-Tumor Activity of Anti-CD30/CEA
#1457 CAR but not Anti-CD25/CEA #1576 CAR

The bispecific #1576 CAR of Seq. ID. No. 18 encoded by the sequence of Seq. ID. No. 17 with specificity for both the lymphocyte activation antigen CD25 and CEA was generated by substituting the HRS3 scFv with the anti-CD25 scFv RFT5 (FIG. 8A; cf FIG. 1). Engineering of T-cells with CAR #1576 was performed as described above by flanking the cDNA for the anti-CD25 scFv with overlapping sequences and sequences for appropriate restriction enzymes by PCR. The sequence of the HRS3 scFv was substituted and the resulting #1576 CAR expressed by transduction with retroviral supernatant according to transduction SOP. The CAR was detected on the T-cell surface by flow cytometry utilizing an antibody against the Fc domain in the extracellular moiety of the CAR. The anti-CD25/CEA CAR was expressed with similar efficiency than monospecific and bispecific anti-CD30 CARs, respectively. Specific CAR T cell activation and lysis of CEA₊ target cells was tested by co-cultivation with CEA₊ and CEA₋ tumor cells respectively. All CARs lysed specifically CEA₊ target cells but the anti-CD30/CEA CAR lysed tumor cells with higher efficiency than monospecific and bispecific anti-CEA and anti-CD25/CEA CARs, respectively (FIG. 8B).

Conclusion

T-cells with the #1457 anti-CD30/CEA CAR but not with the #1576 anti-CD25/CEA CAR were activated against CEA₊ target cells with higher efficiency than anti-CEA-monospecific CAR T cells.

Example 6

Bispecific Anti-CD30 Antibodies with Different
Second Specificity

Bispecific anti-CD30 CAR molecules were generated as described in Example 1 except for substituting the anti-CEA scFv moiety with scFv moieties with specificity against MUC1, C595-scFv, resulting in anti-CD30/MUC1-CAR #1587, and with specificity against TAG-72, B72.3-scFv, resulting in anti-CD30/TAG-72 CAR #1650. The sequences thereof are Seq. Id. Nos. 20 and 19 for #1587 and Seq. ID. Nos. 22 and 21 for #1650.

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acc tgt agt acc agc tcg agt gta agt tac atg cac tgg tac cag cag	528
Thr Cys Ser Thr Ser Ser Ser Val Ser Tyr Met His Trp Tyr Gln Gln	
165 170 175	
aag cca ggt aag gct cca aag ctg ctg atc tac agc aca tcc aac ctg	576
Lys Pro Gly Lys Ala Pro Lys Leu Ile Tyr Ser Thr Ser Asn Leu	
180 185 190	
gct tct ggt gtg cca agc aga ttc agc ggt agc ggt agc ggt acc gac	624
Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp	
195 200 205	
ttc acc ttc acc atc agc agc ctc cag cca gag gac atc gcc acc tac	672
Phe Thr Phe Thr Ile Ser Ser Leu Gln Pro Glu Asp Ile Ala Thr Tyr	
210 215 220	
tac tgc cat cag tgg agt agt tat ccc acg ttc ggc caa ggg acc aag	720
Tyr Cys His Gln Trp Ser Ser Tyr Pro Thr Phe Gly Gln Gly Thr Lys	
225 230 235 240	
gtg gag atc aaa gt	734
Val Glu Ile Lys	

<210> SEQ ID NO 4
 <211> LENGTH: 244
 <212> TYPE: PRT
 <213> ORGANISM: artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 4

Gly Val His Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val
1 5 10 15
Arg Pro Ser Gln Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Thr
20 25 30
Ile Ser Ser Gly Tyr Ser Trp His Trp Val Arg Gln Pro Pro Gly Arg
35 40 45
Gly Leu Glu Trp Ile Gly Tyr Ile Gln Tyr Ser Gly Ile Thr Asn Tyr
50 55 60
Asn Pro Ser Leu Lys Ser Arg Val Thr Met Leu Val Asp Thr Ser Lys
65 70 75 80
Asn Gln Phe Ser Leu Arg Leu Ser Ser Val Thr Ala Ala Asp Thr Ala
85 90 95

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Val Tyr Tyr Cys Ala Arg Glu Asp Tyr Asp Tyr His Trp Tyr Phe Asp
100 105 110

Val Trp Gly Gln Gly Thr Thr Val Thr Val Ser Ser Gly Gly Gly Gly
115 120 125

Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Asp Ile Gln Leu Thr
130 135 140

Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile
145 150 155 160

Thr Cys Ser Thr Ser Ser Ser Val Ser Tyr Met His Trp Tyr Gln Gln
165 170 175

Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile Tyr Ser Thr Ser Asn Leu
180 185 190

Ala Ser Gly Val Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp
195 200 205

Phe Thr Phe Thr Ile Ser Ser Leu Gln Pro Glu Asp Ile Ala Thr Tyr
210 215 220

Tyr Cys His Gln Trp Ser Ser Tyr Pro Thr Phe Gly Gln Gly Thr Lys
225 230 235 240

Val Glu Ile Lys

<210> SEQ ID NO 5
<211> LENGTH: 705
<212> TYPE: DNA
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: synthetic nucleic acid
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(705)

<400> SEQUENCE: 5

gat ccc gcc gag ccc aaa tct cct gac aaa act cac aca tgc cca ccg	48
Asp Pro Ala Glu Pro Lys Ser Pro Asp Lys Thr His Thr Cys Pro Pro	
1 5 10 15	
tgc cca gca cct cca gtc gcg gga ccg tca gtc ttc ctc ttc ccc cca	96
Cys Pro Ala Pro Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro Pro	
20 25 30	
aaa ccc aag gac acc ctc atg atc gcc cgg acc cct gag gtc aca tgc	144
Lys Pro Lys Asp Thr Leu Met Ile Ala Arg Thr Pro Glu Val Thr Cys	
35 40 45	
gtg gtg gtg gac gtg agc cac gaa gac cct gag gtc aag ttc aac tgg	192
Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp	
50 55 60	
tac gtg gac ggc gtg gag gtg cat aat gcc aag aca aag ccg cgg gag	240
Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu	
65 70 75 80	
gag cag tac aac agc acg tac cgt gtg gtc agc gtc ctc acc gtc ctg	288
Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu	
85 90 95	
cac cag gac tgg ctg aat ggc aag gag tac aag tgc aag gtc tcc aac	336
His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn	
100 105 110	
aaa gcc ctc cca gcc ccc atc gag aaa acc atc tcc aaa gcc aaa ggg	384
Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly	
115 120 125	
cag ccc cga gaa cca cag gtg tac acc ctg ccc cca tcc cgg gat gag	432
Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu	
130 135 140	
ctg acc aag aac cag gtc agc ctg acc tgc ctg gtc aaa ggc ttc tat	480

Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	
145					150					155					160	
ccc	agc	gac	atc	gcc	gtg	gag	tgg	gag	agc	aat	ggg	cag	ccg	gag	aac	528
Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	
				165					170					175		
aac	tac	aag	acc	acg	cct	ccc	gtg	ctg	gac	tcc	gac	ggc	tcc	ttc	ttc	576
Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	
			180					185					190			
ctc	tac	agc	aag	ctc	acc	gtg	gac	aag	agc	agg	tgg	cag	cag	ggg	aac	624
Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	
			195				200					205				
gtc	ttc	tca	tgc	tcc	gtg	atg	cat	gag	gct	ctg	cac	aac	cac	tac	acg	672
Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	
			210			215				220						
cag	aag	agc	ctc	tcc	ctg	tct	ccg	ggg	aaa	aaa						705
Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Lys						
225					230					235						

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<210> SEQ ID NO 6
<211> LENGTH: 235
<212> TYPE: PRT
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
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<400> SEQUENCE: 6

Asp	Pro	Ala	Glu	Pro	Lys	Ser	Pro	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro
1				5					10					15	
Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro
			20					25					30		
Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ala	Arg	Thr	Pro	Glu	Val	Thr	Cys
		35					40					45			
Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp
	50					55					60				
Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu
65					70					75					80
Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu
				85					90					95	
His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn
			100					105					110		
Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly
		115					120					125			
Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu
	130					135					140				
Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr
145				150						155					160
Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn
				165					170					175	
Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe
			180					185					190		
Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn
		195					200					205			
Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr
	210					215					220				
Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Lys					
225					230					235					

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<210> SEQ ID NO 7
<211> LENGTH: 204
<212> TYPE: DNA
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: synthetic nucleic acid
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(204)

<400> SEQUENCE: 7

ttt tgg gtg ctg gtg gtg gtt ggt gga gtc ctg gct tgc tat agc ttg      48
Phe Trp Val Leu Val Val Val Gly Gly Val Leu Ala Cys Tyr Ser Leu
1          5          10          15

cta gta aca gtg gcc ttt att att ttc tgg gtg agg agt aag agg agc      96
Leu Val Thr Val Ala Phe Ile Ile Phe Trp Val Arg Ser Lys Arg Ser
          20          25          30

agg ctc ctg cac agt gac tac atg aac atg act ccc cgc cgc ccc ggg      144
Arg Leu Leu His Ser Asp Tyr Met Asn Met Thr Pro Arg Arg Pro Gly
          35          40          45

ccc acc cgc aag cat tac cag gcc tat gcc gcc gca cgc gac ttc gca      192
Pro Thr Arg Lys His Tyr Gln Ala Tyr Ala Ala Ala Arg Asp Phe Ala
          50          55          60

gcc tat cgc tcc      204
Ala Tyr Arg Ser
65

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<210> SEQ ID NO 8
<211> LENGTH: 68
<212> TYPE: PRT
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 8

Phe Trp Val Leu Val Val Val Gly Gly Val Leu Ala Cys Tyr Ser Leu
1          5          10          15

Leu Val Thr Val Ala Phe Ile Ile Phe Trp Val Arg Ser Lys Arg Ser
          20          25          30

Arg Leu Leu His Ser Asp Tyr Met Asn Met Thr Pro Arg Arg Pro Gly
          35          40          45

Pro Thr Arg Lys His Tyr Gln Ala Tyr Ala Ala Ala Arg Asp Phe Ala
          50          55          60

Ala Tyr Arg Ser
65

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<210> SEQ ID NO 9
<211> LENGTH: 414
<212> TYPE: DNA
<213> ORGANISM: human
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(414)

<400> SEQUENCE: 9

gat ccc aaa ctc tgc tac ctg ctg gat gga atc ctc ttc atc tat ggt      48
Asp Pro Lys Leu Cys Tyr Leu Leu Asp Gly Ile Leu Phe Ile Tyr Gly
1          5          10          15

gtc att ctc act gcc ttg ttc ctg aga gtg aag ttc agc agg agc gca      96
Val Ile Leu Thr Ala Leu Phe Leu Arg Val Lys Phe Ser Arg Ser Ala
          20          25          30

gac gcc ccc gcg tac cag cag gcc cag aac cag ctc tat aac gag ctc      144
Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu
          35          40          45

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aat cta gga cga aga gag gag tac gat gtt ttg gac aag aga cgt ggc      192
Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly
  50                      55                      60

cgg gac cct gag atg ggg gga aag ccg cag aga agg aag aac cct cag      240
Arg Asp Pro Glu Met Gly Gly Lys Pro Gln Arg Arg Lys Asn Pro Gln
  65                      70                      75                      80

gaa ggc ctg tac aat gaa ctg cag aaa gat aag atg gcg gag gcc tac      288
Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr
                      85                      90                      95

agt gag att ggg atg aaa ggc gag cgc cgg agg ggc aag ggg cac gat      336
Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp
                      100                      105                      110

ggc ctt tac cag ggt ctc agt aca gcc acc aag gac acc tac gac gcc      384
Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala
                      115                      120                      125

ctt cac atg cag gcc ctg ccc cct cgc taa                                414
Leu His Met Gln Ala Leu Pro Pro Arg
                      130                      135

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<210> SEQ ID NO 10
<211> LENGTH: 137
<212> TYPE: PRT
<213> ORGANISM: human

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<400> SEQUENCE: 10

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Asp Pro Lys Leu Cys Tyr Leu Leu Asp Gly Ile Leu Phe Ile Tyr Gly
  1                      5                      10                      15

Val Ile Leu Thr Ala Leu Phe Leu Arg Val Lys Phe Ser Arg Ser Ala
                      20                      25                      30

Asp Ala Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu
                      35                      40                      45

Asn Leu Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly
                      50                      55                      60

Arg Asp Pro Glu Met Gly Gly Lys Pro Gln Arg Arg Lys Asn Pro Gln
                      65                      70                      75                      80

Glu Gly Leu Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr
                      85                      90                      95

Ser Glu Ile Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp
                      100                      105                      110

Gly Leu Tyr Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala
                      115                      120                      125

Leu His Met Gln Ala Leu Pro Pro Arg
                      130                      135

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<210> SEQ ID NO 11
<211> LENGTH: 201
<212> TYPE: DNA
<213> ORGANISM: human
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(201)

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<400> SEQUENCE: 11

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gat ccc aaa ctc tgc tat atc ctg gat gcc atc ctg ttt ctg tat gga      48
Asp Pro Lys Leu Cys Tyr Ile Leu Asp Ala Ile Leu Phe Leu Tyr Gly
  1                      5                      10                      15

att gtc ctc acc ctc ctc tac tgt cga ctg aag atc caa gtg cga aag      96
Ile Val Leu Thr Leu Leu Tyr Cys Arg Leu Lys Ile Gln Val Arg Lys
                      20                      25                      30

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gca gct ata acc agc tat gag aaa tca gat ggt gtt tac acg ggc ctg      144
Ala Ala Ile Thr Ser Tyr Glu Lys Ser Asp Gly Val Tyr Thr Gly Leu
      35                      40                      45

agc acc agg aac cag gag act tac gag act ctg aag cat gag aaa cca      192
Ser Thr Arg Asn Gln Glu Thr Tyr Glu Thr Leu Lys His Glu Lys Pro
      50                      55                      60

cca cag tag      201
Pro Gln
65

<210> SEQ ID NO 12
<211> LENGTH: 66
<212> TYPE: PRT
<213> ORGANISM: human

<400> SEQUENCE: 12

Asp Pro Lys Leu Cys Tyr Ile Leu Asp Ala Ile Leu Phe Leu Tyr Gly
1          5          10          15
Ile Val Leu Thr Leu Leu Tyr Cys Arg Leu Lys Ile Gln Val Arg Lys
      20          25          30
Ala Ala Ile Thr Ser Tyr Glu Lys Ser Asp Gly Val Tyr Thr Gly Leu
      35          40          45
Ser Thr Arg Asn Gln Glu Thr Tyr Glu Thr Leu Lys His Glu Lys Pro
      50          55          60
Pro Gln
65

<210> SEQ ID NO 13
<211> LENGTH: 2943
<212> TYPE: DNA
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: synthetic sequence
<220> FEATURE:
<221> NAME/KEY: CDS
<222> LOCATION: (1)..(2943)

<400> SEQUENCE: 13

ctc tag act gcc atg gat ttt cag gtg cag att ttc agc ttc ctg cta      48
Leu Thr Ala Met Asp Phe Gln Val Gln Ile Phe Ser Phe Leu Leu
1          5          10          15

atc agt gcc tca gtc ata atg tct aga atg gcc cag gtg caa ctg cag      96
Ile Ser Ala Ser Val Ile Met Ser Arg Met Ala Gln Val Gln Leu Gln
      20          25          30

cag tca ggg gct gag ctg gct aga cct ggg gct tca gtg aag atg tcc      144
Gln Ser Gly Ala Glu Leu Ala Arg Pro Gly Ala Ser Val Lys Met Ser
      35          40          45

tgc aag gct tct ggc tac acc ttt act acc tac aca ata cac tgg gta      192
Cys Lys Ala Ser Gly Tyr Thr Phe Thr Thr Tyr Thr Ile His Trp Val
      50          55          60

aga cgg agg cct gga cac gat ctg gaa tgg att gga tac att aat cct      240
Arg Arg Arg Pro Gly His Asp Leu Glu Trp Ile Gly Tyr Ile Asn Pro
      65          70          75

agc agt gga tgt tct gac tac aat caa aac ttc aag ggc aag acc aca      288
Ser Ser Gly Cys Ser Asp Tyr Asn Gln Asn Phe Lys Gly Lys Thr Thr
      80          85          90          95

ttg act gca gac aag tcc tcc aac aca gcc tac atg caa ctg aac agc      336
Leu Thr Ala Asp Lys Ser Ser Asn Thr Ala Tyr Met Gln Leu Asn Ser
      100          105          110

ctg aca tct gag gac tct gcg gtc tat tac tgt gca aga aga gcg gac      384
Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys Ala Arg Arg Ala Asp
      115          120          125

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tat ggt aac tac gaa tat acc tgg ttt gct tac tgg ggc caa ggg acc Tyr Gly Asn Tyr Glu Tyr Thr Trp Phe Ala Tyr Trp Gly Gln Gly Thr 130 135 140	432
acg gtc acc gtc tcc tca agt gga ggc ggt tca ggt gga ggt ggc tct Thr Val Thr Val Ser Ser Ser Gly Gly Gly Ser Gly Gly Gly Ser 145 150 155	480
ggc ggt ggc gga tcg gtc atc gag ctc act cag tct cca aaa ttc atg Gly Gly Gly Gly Ser Val Ile Glu Leu Thr Gln Ser Pro Lys Phe Met 160 165 170 175	528
tcc aca tca gta gga gac agg gtc aac gtc acc tac aag gcc agt cag Ser Thr Ser Val Gly Asp Arg Val Asn Val Thr Tyr Lys Ala Ser Gln 180 185 190	576
aat gtg ggt act aat gta gcc tgg ttt caa caa aaa cca ggg caa tct Asn Val Gly Thr Asn Val Ala Trp Phe Gln Gln Lys Pro Gly Gln Ser 195 200 205	624
cct aaa gtt ctg att tac tcg gca tct tac cga tac agt gga gtc cct Pro Lys Val Leu Ile Tyr Ser Ala Ser Tyr Arg Tyr Ser Gly Val Pro 210 215 220	672
gat cgc ttc aca ggc agt gga tct gga aca gat ttc act ctc acc atc Asp Arg Phe Thr Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile 225 230 235	720
agc aat gtg cag tct gaa gac ttg gca gag tat ttc tgt cag caa tat Ser Asn Val Gln Ser Glu Asp Leu Ala Glu Tyr Phe Cys Gln Gln Tyr 240 245 250 255	768
cac acc tat cct ctc acg ttc gga ggg ggc acc aag ctg gaa atc aaa His Thr Tyr Pro Leu Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys 260 265 270	816
cgg tca gat ccc gga ggc ggg ggt tct ggt gga ggc gga agc ggt ggc Arg Ser Asp Pro Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly 275 280 285	864
ggg ggt tca gga ggc ggg ggt tcc ggt ggt ggc ggc agt ggt ggt ggc Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly 290 295 300	912
ggc agt ggt ggt ggc ggc agt ggt ggt ggc ggc agt ggt gtc cac tcc Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Val His Ser 305 310 315	960
cag gtc caa ctg cag gag tca ggt cca ggt ctt gtg aga cct agc cag Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Arg Pro Ser Gln 320 325 330 335	1008
acc ctg agc ctg acc tgc acc gtg tct ggc ttc acc atc agc agt ggt Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Thr Ile Ser Ser Gly 340 345 350	1056
tat agc tgg cac tgg gtg aga cag cca cct gga cga ggt ctt gag tgg Tyr Ser Trp His Trp Val Arg Gln Pro Pro Gly Arg Gly Leu Glu Trp 355 360 365	1104
att gga tac ata cag tac agt ggt atc act aac tac aac ccc tct ctc Ile Gly Tyr Ile Gln Tyr Ser Gly Ile Thr Asn Tyr Asn Pro Ser Leu 370 375 380	1152
aaa agt aga gtg aca atg ctg gta gac acc agc aag aac cag ttc agc Lys Ser Arg Val Thr Met Leu Val Asp Thr Ser Lys Asn Gln Phe Ser 385 390 395	1200
ctg aga ctc agc agc gtg aca gcc gcc gac acc gcg gtc tat tat tgt Leu Arg Leu Ser Ser Val Thr Ala Ala Asp Thr Ala Val Tyr Tyr Cys 400 405 410 415	1248
gca aga gaa gac tat gat tac cac tgg tac ttc gat gtc tgg ggc caa Ala Arg Glu Asp Tyr Asp Tyr His Trp Tyr Phe Asp Val Trp Gly Gln 420 425 430	1296
ggg acc acg gtc acc gtc tcc tca gga ggt ggt gga tcg ggc ggt ggc Gly Thr Thr Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly 1344	

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435	440	445	
ggg tcg ggt ggc ggc gga tct gac atc cag ctg acc cag agc cca agc Gly Ser Gly Gly Gly Gly Ser Asp Ile Gln Leu Thr Gln Ser Pro Ser 450 455 460			1392
agc ctg agc gcc agc gtg ggt gac aga gtg acc atc acc tgt agt acc Ser Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Ser Thr 465 470 475			1440
agc tcg agt gta agt tac atg cac tgg tac cag cag aag cca ggt aag Ser Ser Ser Val Ser Tyr Met His Trp Tyr Gln Gln Lys Pro Gly Lys 480 485 490 495			1488
gct cca aag ctg ctg atc tac agc aca tcc aac ctg gct tct ggt gtg Ala Pro Lys Leu Ile Tyr Ser Thr Ser Asn Leu Ala Ser Gly Val 500 505 510			1536
cca agc aga ttc agc ggt agc ggt agc ggt acc gac ttc acc ttc acc Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Phe Thr 515 520 525			1584
atc agc agc ctc cag cca gag gac atc gcc acc tac tac tgc cat cag Ile Ser Ser Leu Gln Pro Glu Asp Ile Ala Thr Tyr Tyr Cys His Gln 530 535 540			1632
tgg agt agt tat ccc acg ttc ggc caa ggg acc aag gtg gag atc aaa Trp Ser Ser Tyr Pro Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys 545 550 555			1680
gtg gat ccc gcc gag ccc aaa tct cct gac aaa act cac aca tgc cca Val Asp Pro Ala Glu Pro Lys Ser Pro Asp Lys Thr His Thr Cys Pro 560 565 570 575			1728
ccg tgc cca gca cct cca gtc gcg gga ccg tca gtc ttc ctc ttc ccc Pro Cys Pro Ala Pro Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro 580 585 590			1776
cca aaa ccc aag gac acc ctc atg atc gcc cgg acc cct gag gtc aca Pro Lys Pro Lys Asp Thr Leu Met Ile Ala Arg Thr Pro Glu Val Thr 595 600 605			1824
tgc gtg gtg gtg gac gtg agc cac gaa gac cct gag gtc aag ttc aac Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn 610 615 620			1872
tgg tac gtg gac ggc gtg gag gtg cat aat gcc aag aca aag ccg cgg Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg 625 630 635			1920
gag gag cag tac aac agc acg tac cgt gtg gtc agc gtc ctc acc gtc Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val 640 645 650 655			1968
ctg cac cag gac tgg ctg aat ggc aag gag tac aag tgc aag gtc tcc Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser 660 665 670			2016
aac aaa gcc ctc cca gcc ccc atc gag aaa acc atc tcc aaa gcc aaa Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys 675 680 685			2064
ggg cag ccc cga gaa cca cag gtg tac acc ctg ccc cca tcc cgg gat Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp 690 695 700			2112
gag ctg acc aag aac cag gtc agc ctg acc tgc ctg gtc aaa ggc ttc Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe 705 710 715			2160
tat ccc agc gac atc gcc gtg gag tgg gag agc aat ggg cag ccg gag Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu 720 725 730 735			2208
aac aac tac aag acc acg cct ccc gtg ctg gac tcc gac ggc tcc ttc Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe 740 745 750			2256
ttc ctc tac agc aag ctc acc gtg gac aag agc agg tgg cag cag ggg			2304

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Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly		
			755					760					765				
aac	gtc	ttc	tca	tgc	tcc	gtg	atg	cat	gag	gct	ctg	cac	aac	cac	tac	2352	
Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr		
		770					775					780					
acg	cag	aag	agc	ctc	tcc	ctg	tct	ccg	ggt	aaa	aaa	gat	ccc	aaa	ttt	2400	
Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Lys	Asp	Pro	Lys	Phe		
	785					790					795						
tgg	gtg	ctg	gtg	gtg	gtt	ggt	gga	gtc	ctg	gct	tgc	tat	agc	ttg	cta	2448	
Trp	Val	Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	Leu		
	800				805					810				815			
gta	aca	gtg	gcc	ttt	att	att	ttc	tgg	gtg	agg	agt	aag	agg	agc	agg	2496	
Val	Thr	Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Arg	Ser	Lys	Arg	Ser	Arg		
			820					825					830				
ctc	ctg	cac	agt	gac	tac	atg	aac	atg	act	ccc	cgc	cgc	ccc	ggg	ccc	2544	
Leu	Leu	His	Ser	Asp	Tyr	Met	Asn	Met	Thr	Pro	Arg	Arg	Pro	Gly	Pro		
			835					840					845				
acc	cgc	aag	cat	tac	cag	gcc	tat	gcc	gcc	gca	cgc	gac	ttc	gca	gcc	2592	
Thr	Arg	Lys	His	Tyr	Gln	Ala	Tyr	Ala	Ala	Ala	Arg	Asp	Phe	Ala	Ala		
		850				855						860					
tat	cgc	tcc	ctg	aga	gtg	aag	ttc	agc	agg	agc	gca	gac	gcc	ccc	gcg	2640	
Tyr	Arg	Ser	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala		
	865					870					875						
tac	cag	cag	ggc	cag	aac	cag	ctc	tat	aac	gag	ctc	aat	cta	gga	cga	2688	
Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg		
	880				885					890					895		
aga	gag	gag	tac	gat	gtt	ttg	gac	aag	aga	cgt	ggc	cgg	gac	cct	gag	2736	
Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu		
			900						905					910			
atg	ggg	gga	aag	ccg	aga	agg	aag	aac	cct	cag	gaa	ggc	ctg	tac	aat	2784	
Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn		
			915					920					925				
gaa	ctg	cag	aaa	gat	aag	atg	gcg	gag	gcc	tac	agt	gag	att	ggg	atg	2832	
Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met		
		930				935						940					
aaa	ggc	gag	cgc	cgg	agg	ggc	aag	ggg	cac	gat	ggc	ctt	tac	cag	ggt	2880	
Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly		
	945					950					955						
ctc	agt	aca	gcc	acc	aag	gac	acc	tac	gac	gcc	ctt	cac	atg	cag	gcc	2928	
Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	Gln	Ala		
	960				965					970				975			
ctg	ccc	cct	cgc	taa												2943	
Leu	Pro	Pro	Arg														

<210> SEQ ID NO 14
 <211> LENGTH: 978
 <212> TYPE: PRT
 <213> ORGANISM: artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 14

Thr	Ala	Met	Asp	Phe	Gln	Val	Gln	Ile	Phe	Ser	Phe	Leu	Leu	Ile	Ser
1				5					10					15	
Ala	Ser	Val	Ile	Met	Ser	Arg	Met	Ala	Gln	Val	Gln	Leu	Gln	Gln	Ser
		20						25				30			
Gly	Ala	Glu	Leu	Ala	Arg	Pro	Gly	Ala	Ser	Val	Lys	Met	Ser	Cys	Lys
	35						40				45				
Ala	Ser	Gly	Tyr	Thr	Phe	Thr	Thr	Tyr	Thr	Ile	His	Trp	Val	Arg	Arg
	50					55				60					

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Arg	Pro	Gly	His	Asp	Leu	Glu	Trp	Ile	Gly	Tyr	Ile	Asn	Pro	Ser	Ser	
65					70					75					80	
Gly	Cys	Ser	Asp	Tyr	Asn	Gln	Asn	Phe	Lys	Gly	Lys	Thr	Thr	Leu	Thr	
			85						90					95		
Ala	Asp	Lys	Ser	Ser	Asn	Thr	Ala	Tyr	Met	Gln	Leu	Asn	Ser	Leu	Thr	
			100					105					110			
Ser	Glu	Asp	Ser	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Arg	Ala	Asp	Tyr	Gly	
	115						120					125				
Asn	Tyr	Glu	Tyr	Thr	Trp	Phe	Ala	Tyr	Trp	Gly	Gln	Gly	Thr	Thr	Val	
	130					135						140				
Thr	Val	Ser	Ser	Ser	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	
145					150					155					160	
Gly	Gly	Ser	Val	Ile	Glu	Leu	Thr	Gln	Ser	Pro	Lys	Phe	Met	Ser	Thr	
			165						170					175		
Ser	Val	Gly	Asp	Arg	Val	Asn	Val	Thr	Tyr	Lys	Ala	Ser	Gln	Asn	Val	
		180						185					190			
Gly	Thr	Asn	Val	Ala	Trp	Phe	Gln	Gln	Lys	Pro	Gly	Gln	Ser	Pro	Lys	
		195					200					205				
Val	Leu	Ile	Tyr	Ser	Ala	Ser	Tyr	Arg	Tyr	Ser	Gly	Val	Pro	Asp	Arg	
	210					215					220					
Phe	Thr	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Asn	
225					230					235					240	
Val	Gln	Ser	Glu	Asp	Leu	Ala	Glu	Tyr	Phe	Cys	Gln	Gln	Tyr	His	Thr	
			245						250					255		
Tyr	Pro	Leu	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Arg	Ser	
		260						265					270			
Asp	Pro	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	
		275					280					285				
Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	
	290					295					300					
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Val	His	Ser	Gln	Val	
305					310					315					320	
Gln	Leu	Gln	Glu	Ser	Gly	Pro	Gly	Leu	Val	Arg	Pro	Ser	Gln	Thr	Leu	
			325					330						335		
Ser	Leu	Thr	Cys	Thr	Val	Ser	Gly	Phe	Thr	Ile	Ser	Ser	Gly	Tyr	Ser	
		340						345					350			
Trp	His	Trp	Val	Arg	Gln	Pro	Pro	Gly	Arg	Gly	Leu	Glu	Trp	Ile	Gly	
		355				360					365					
Tyr	Ile	Gln	Tyr	Ser	Gly	Ile	Thr	Asn	Tyr	Asn	Pro	Ser	Leu	Lys	Ser	
	370					375					380					
Arg	Val	Thr	Met	Leu	Val	Asp	Thr	Ser	Lys	Asn	Gln	Phe	Ser	Leu	Arg	
385					390					395					400	
Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	
			405						410				415			
Glu	Asp	Tyr	Asp	Tyr	His	Trp	Tyr	Phe	Asp	Val	Trp	Gly	Gln	Gly	Thr	
		420						425					430			
Thr	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	
	435					440						445				
Gly	Gly	Gly	Gly	Ser	Asp	Ile	Gln	Leu	Thr	Gln	Ser	Pro	Ser	Ser	Leu	
	450					455					460					
Ser	Ala	Ser	Val	Gly	Asp	Arg	Val	Thr	Ile	Thr	Cys	Ser	Thr	Ser	Ser	
465					470					475					480	

Ser	Val	Ser	Tyr	Met	His	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ala	Pro	
				485					490							495
Lys	Leu	Leu	Ile	Tyr	Ser	Thr	Ser	Asn	Leu	Ala	Ser	Gly	Val	Pro	Ser	
				500					505							510
Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Phe	Thr	Ile	Ser	
				515					520							525
Ser	Leu	Gln	Pro	Glu	Asp	Ile	Ala	Thr	Tyr	Tyr	Cys	His	Gln	Trp	Ser	
				530					535							540
Ser	Tyr	Pro	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys	Val	Asp	
				545					550							555
Pro	Ala	Glu	Pro	Lys	Ser	Pro	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	
				565					570							575
Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	
				580					585							590
Pro	Lys	Asp	Thr	Leu	Met	Ile	Ala	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	
				595					600							605
Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	
				610					615							620
Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	
				625					630							635
Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	
				645					650							655
Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	
				660					665							670
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	
				675					680							685
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	
				690					695							700
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	
				705					710							715
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	
				725					730							735
Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	
				740					745							750
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	
				755					760							765
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	
				770					775							780
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Lys	Asp	Pro	Lys	Phe	Trp	Val	
				785					790							795
Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	Leu	Val	Thr	
				805					810							815
Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Arg	Ser	Lys	Arg	Ser	Arg	Leu	Leu	
				820					825							830
His	Ser	Asp	Tyr	Met	Asn	Met	Thr	Pro	Arg	Arg	Pro	Gly	Pro	Thr	Arg	
				835					840							845
Lys	His	Tyr	Gln	Ala	Tyr	Ala	Ala	Ala	Arg	Asp	Phe	Ala	Ala	Tyr	Arg	
				850					855							860
Ser	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln	
				865					870							875
Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu	
				885					890							895
Glu	Tyr	Asp	Val	Leu	Asp	Lys										

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900	905	910
Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu		
915	920	925
Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly		
930	935	940
Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser		
945	950	955
960		
Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro		
965	970	975

Pro Arg

<210> SEQ ID NO 15
 <211> LENGTH: 25
 <212> TYPE: PRT
 <213> ORGANISM: artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: synthetic peptide
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (8)..(20)
 <223> OTHER INFORMATION: SVFLFPPKPKDTL

<400> SEQUENCE: 15

Pro	Glu	Leu	Leu	Gly	Gly	Pro	Xaa	Xaa	Xaa	Xaa	Xaa	Xaa	Xaa	Xaa	Xaa
1			5					10						15	

Xaa	Xaa	Xaa	Xaa	Met	Ile	Ser	Arg	Thr
			20				25	

<210> SEQ ID NO 16
 <211> LENGTH: 24
 <212> TYPE: PRT
 <213> ORGANISM: artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: synthetic peptide
 <220> FEATURE:
 <221> NAME/KEY: misc_feature
 <222> LOCATION: (7)..(19)
 <223> OTHER INFORMATION: SVFLFPPKPKDTL

<400> SEQUENCE: 16

Pro	Pro	Val	Ala	Gly	Pro	Xaa	Xaa	Xaa	Xaa	Xaa	Xaa	Xaa	Xaa	Xaa	Xaa
1			5				10						15		

Xaa	Xaa	Xaa	Met	Ile	Ala	Arg	Thr
			20				

<210> SEQ ID NO 17
 <211> LENGTH: 2901
 <212> TYPE: DNA
 <213> ORGANISM: artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: synthetic molecule
 <220> FEATURE:
 <221> NAME/KEY: CDS
 <222> LOCATION: (1)..(2901)

<400> SEQUENCE: 17

atg	gat	ttt	cag	gtg	cag	att	ttc	agc	ttc	ctg	cta	atc	agt	gcc	tca	48
Met	Asp	Phe	Gln	Val	Gln	Ile	Phe	Ser	Phe	Leu	Leu	Ile	Ser	Ala	Ser	
1			5					10						15		

gtc	ata	atg	tct	aga	cag	gtg	aag	ctg	cag	cag	tct	ggg	act	gtg	ctg	96
Val	Ile	Met	Ser	Arg	Gln	Val	Lys	Leu	Gln	Gln	Ser	Gly	Thr	Val	Leu	
			20					25					30			

gca	agg	cct	ggg	gct	tcc	gtg	aag	atg	tcc	tgc	aag	gct	tct	ggc	tac	144
Ala	Arg	Pro	Gly	Ala	Ser	Val	Lys	Met	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	

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35	40	45	
agg ttt acc aac tac tgg atg cac tgg gta aaa cag agg cct gga cag			192
Arg Phe Thr Asn Tyr Trp Met His Trp Val Lys Gln Arg Pro Gly Gln			
50	55	60	
ggt cta gaa tgg att ggt gtt att tat cct gga aat agt gat act agc			240
Gly Leu Glu Trp Ile Gly Val Ile Tyr Pro Gly Asn Ser Asp Thr Ser			
65	70	75	80
tac aac cag aag ttc aag ggc aag gcc aaa ctg act gca gtc aca tcc			288
Tyr Asn Gln Lys Phe Lys Gly Lys Ala Lys Leu Thr Ala Val Thr Ser			
85	90	95	
gcc agc act gcc tac atg gag ctc agc agc ctg aca aat gag gac tct			336
Ala Ser Thr Ala Tyr Met Glu Leu Ser Ser Leu Thr Asn Glu Asp Ser			
100	105	110	
gcg gtc tat tac tgt aca aga gag gga gaa ggc tct gac tac tgg ggc			384
Ala Val Tyr Tyr Cys Thr Arg Glu Gly Glu Gly Ser Asp Tyr Trp Gly			
115	120	125	
caa ggg acc acg gtc acc gtc tcc tca ggt gga ggc ggt tca ggc gga			432
Gln Gly Thr Thr Val Thr Val Ser Ser Gly Gly Gly Ser Gly Gly			
130	135	140	
ggt ggc tct ggc ggt ggc gga tcg caa att gtt ctc acc cag tct cca			480
Gly Gly Ser Gly Gly Gly Gly Ser Gln Ile Val Leu Thr Gln Ser Pro			
145	150	155	160
gca acc atg gct gca tct ccc ggg gag aag atc act atc acc tgc agt			528
Ala Thr Met Ala Ala Ser Pro Gly Glu Lys Ile Thr Ile Thr Cys Ser			
165	170	175	
gcc agc tca agt ata agt tcc aat tac ttg cat tgg tat cag cag aag			576
Ala Ser Ser Ser Ile Ser Ser Asn Tyr Leu His Trp Tyr Gln Gln Lys			
180	185	190	
cca gga ttc tcc cct aaa ctc ttg att tat agg act tcc aat ctg gct			624
Pro Gly Phe Ser Pro Lys Leu Leu Ile Tyr Arg Thr Ser Asn Leu Ala			
195	200	205	
tct gga gtc cca gct cgc ttc agt ggc agt ggg tct ggg acc tct tac			672
Ser Gly Val Pro Ala Arg Phe Ser Gly Ser Gly Ser Gly Thr Ser Tyr			
210	215	220	
tct ctc aca att ggc acc atg gag gct gaa gat gtt gcc act tac tac			720
Ser Leu Thr Ile Gly Thr Met Glu Ala Glu Asp Val Ala Thr Tyr Tyr			
225	230	235	240
tgc cag cag ggt agt agt ata ccg tac acg ttc gga ggg ggg acc aag			768
Cys Gln Gln Gly Ser Ser Ile Pro Tyr Thr Phe Gly Gly Gly Thr Lys			
245	250	255	
ctg gag ctg gcg gcc gca gga ggc ggg ggt tct ggt gga ggc gga agc			816
Leu Glu Leu Ala Ala Ala Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser			
260	265	270	
ggt ggc ggg ggt tca gga ggc ggg ggt tcc ggt ggt ggc ggc agt ggt			864
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly			
275	280	285	
ggt ggc ggc agt ggt ggt ggc ggc agt ggt ggt ggc ggc agt ggt gtc			912
Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Val			
290	295	300	
cac tcc cag gtc caa ctg cag gag tca ggt cca ggt ctt gtg aga cct			960
His Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Arg Pro			
305	310	315	320
agc cag acc ctg agc ctg acc tgc acc gtg tct ggc ttc acc atc agc			1008
Ser Gln Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Thr Ile Ser			
325	330	335	
agt ggt tat agc tgg cac tgg gtg aga cag cca cct gga cga ggt ctt			1056
Ser Gly Tyr Ser Trp His Trp Val Arg Gln Pro Pro Gly Arg Gly Leu			
340	345	350	
gag tgg att gga tac ata cag tac agt ggt atc act aac tac aac ccc			1104

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Glu	Trp	Ile	Gly	Tyr	Ile	Gln	Tyr	Ser	Gly	Ile	Thr	Asn	Tyr	Asn	Pro	
		355					360					365				
tct	ctc	aaa	agt	aga	gtg	aca	atg	ctg	gta	gac	acc	agc	aag	aac	cag	1152
Ser	Leu	Lys	Ser	Arg	Val	Thr	Met	Leu	Val	Asp	Thr	Ser	Lys	Asn	Gln	
	370					375					380					
ttc	agc	ctg	aga	ctc	agc	agc	gtg	aca	gcc	gcc	gac	acc	gcg	gtc	tat	1200
Phe	Ser	Leu	Arg	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	
385					390					395					400	
tat	tgt	gca	aga	gaa	gac	tat	gat	tac	cac	tgg	tac	ttc	gat	gtc	tgg	1248
Tyr	Cys	Ala	Arg	Glu	Asp	Tyr	Asp	Tyr	His	Trp	Tyr	Phe	Asp	Val	Trp	
			405						410					415		
ggc	caa	ggg	acc	acg	gtc	acc	gtc	tcc	tca	gga	ggg	ggg	gga	tcg	ggc	1296
Gly	Gln	Gly	Thr	Thr	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	
			420					425					430			
ggg	ggc	ggg	tcg	ggg	ggc	ggc	gga	tct	gac	atc	cag	ctg	acc	cag	agc	1344
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Ile	Gln	Leu	Thr	Gln	Ser	
	435					440					445					
cca	agc	agc	ctg	agc	gcc	agc	gtg	ggg	gac	aga	gtg	acc	atc	acc	tgt	1392
Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly	Asp	Arg	Val	Thr	Ile	Thr	Cys	
	450					455					460					
agt	acc	agc	tcg	agt	gta	agt	tac	atg	cac	tgg	tac	cag	cag	aag	cca	1440
Ser	Thr	Ser	Ser	Ser	Val	Ser	Tyr	Met	His	Trp	Tyr	Gln	Gln	Lys	Pro	
465					470					475					480	
ggg	aag	gct	cca	aag	ctg	ctg	atc	tac	agc	aca	tcc	aac	ctg	gct	tct	1488
Gly	Lys	Ala	Pro	Lys	Leu	Leu	Ile	Tyr	Ser	Thr	Ser	Asn	Leu	Ala	Ser	
			485					490						495		
ggg	gtg	cca	agc	aga	ttc	agc	ggg	agc	ggg	agc	ggg	acc	gac	ttc	acc	1536
Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	
		500					505					510				
ttc	acc	atc	agc	agc	ctc	cag	cca	gag	gac	atc	gcc	acc	tac	tac	tgc	1584
Phe	Thr	Ile	Ser	Ser	Leu	Gln	Pro	Glu	Asp	Ile	Ala	Thr	Tyr	Tyr	Cys	
	515					520					525					
cat	cag	tgg	agt	agt	tat	ccc	acg	ttc	ggc	caa	ggg	acc	aag	gtg	gag	1632
His	Gln	Trp	Ser	Ser	Tyr	Pro	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	
	530					535					540					
atc	aaa	gtg	gat	ccc	gcc	gag	ccc	aaa	tct	cct	gac	aaa	act	cac	aca	1680
Ile	Lys	Val	Asp	Pro	Ala	Glu	Pro	Lys	Ser	Pro	Asp	Lys	Thr	His	Thr	
545					550					555					560	
tgc	cca	ccg	tgc	cca	gca	cct	cca	gtc	gcg	gga	ccg	tca	gtc	ttc	ctc	1728
Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	
			565						570					575		
ttc	ccc	cca	aaa	ccc	aag	gac	acc	ctc	atg	atc	gcc	egg	acc	cct	gag	1776
Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ala	Arg	Thr	Pro	Glu	
		580					585						590			
gtc	aca	tgc	gtg	gtg	gtg	gac	gtg	agc	cac	gaa	gac	cct	gag	gtc	aag	1824
Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	
	595					600						605				
ttc	aac	tgg	tac	gtg	gac	ggc	gtg	gag	gtg	cat	aat	gcc	aag	aca	aag	1872
Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	
	610					615					620					
ccg	egg	gag	gag	cag	tac	aac	agc	acg	tac	cgt	gtg	gtc	agc	gtc	ctc	1920
Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	
625					630					635					640	
acc	gtc	ctg	cac	cag	gac	tgg	ctg	aat	ggc	aag	gag	tac	aag	tgc	aag	1968
Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	
			645						650					655		
gtc	tcc	aac	aaa	gcc	ctc	cca	gcc	ccc	atc	gag	aaa	acc	atc	tcc	aaa	2016
Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	
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gcc aaa ggg cag ccc cga gaa cca cag gtg tac acc ctg ccc cca tcc	2064
Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser	
675 680 685	
cgg gat gag ctg acc aag aac cag gtc agc ctg acc tgc ctg gtc aaa	2112
Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys	
690 695 700	
ggc ttc tat ccc agc gac atc gcc gtg gag tgg gag agc aat ggg cag	2160
Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln	
705 710 715 720	
ccg gag aac aac tac aag acc acg cct ccc gtg ctg gac tcc gac ggc	2208
Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly	
725 730 735	
tcc ttc ttc ctc tac agc aag ctc acc gtg gac aag agc agg tgg cag	2256
Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln	
740 745 750	
cag ggg aac gtc ttc tca tgc tcc gtg atg cat gag gct ctg cac aac	2304
Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn	
755 760 765	
cac tac acg cag aag agc ctc tcc ctg tct ccg ggt aaa aaa gat ccc	2352
His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys Lys Asp Pro	
770 775 780	
aaa ttt tgg gtg ctg gtg gtg gtt ggt gga gtc ctg gct tgc tat agc	2400
Lys Phe Trp Val Leu Val Val Val Gly Gly Val Leu Ala Cys Tyr Ser	
785 790 795 800	
ttg cta gta aca gtg gcc ttt att att ttc tgg gtg agg agt aag agg	2448
Leu Leu Val Thr Val Ala Phe Ile Ile Phe Trp Val Arg Ser Lys Arg	
805 810 815	
agc agg ctc ctg cac agt gac tac atg aac atg act ccc cgc cgc ccc	2496
Ser Arg Leu Leu His Ser Asp Tyr Met Asn Met Thr Pro Arg Arg Pro	
820 825 830	
ggg ccc acc cgc aag cat tac cag gcc tat gcc gcc gca cgc gac ttc	2544
Gly Pro Thr Arg Lys His Tyr Gln Ala Tyr Ala Ala Ala Arg Asp Phe	
835 840 845	
gca gcc tat cgc tcc ctg aga gtg aag ttc agc agg agc gca gac gcc	2592
Ala Ala Tyr Arg Ser Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala	
850 855 860	
ccc gcg tac cag cag ggc cag aac cag ctc tat aac gag ctc aat cta	2640
Pro Ala Tyr Gln Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu	
865 870 875 880	
gga cga aga gag gag tac gat gtt ttg gac aag aga cgt ggc cgg gac	2688
Gly Arg Arg Glu Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp	
885 890 895	
cct gag atg ggg gga aag ccg aga agg aag aac cct cag gaa ggc ctg	2736
Pro Glu Met Gly Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu	
900 905 910	
tac aat gaa ctg cag aaa gat aag atg gcg gag gcc tac agt gag att	2784
Tyr Asn Glu Leu Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile	
915 920 925	
ggg atg aaa ggc gag cgc cgg agg ggc aag ggg cac gat ggc ctt tac	2832
Gly Met Lys Gly Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr	
930 935 940	
cag ggt ctc agt aca gcc acc aag gac acc tac gac gcc ctt cac atg	2880
Gln Gly Leu Ser Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met	
945 950 955 960	
cag gcc ctg ccc cct cgc taa	2901
Gln Ala Leu Pro Pro Arg	
965	

<210> SEQ ID NO 18

<211> LENGTH: 966

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<212> TYPE: PRT
<213> ORGANISM: artificial
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 18

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Val Ile Met Ser Arg Gln Val Lys Leu Gln Gln Ser Gly Thr Val Leu
 20             25             30

Ala Arg Pro Gly Ala Ser Val Lys Met Ser Cys Lys Ala Ser Gly Tyr
 35             40             45

Arg Phe Thr Asn Tyr Trp Met His Trp Val Lys Gln Arg Pro Gly Gln
 50             55             60

Gly Leu Glu Trp Ile Gly Val Ile Tyr Pro Gly Asn Ser Asp Thr Ser
 65             70             75             80

Tyr Asn Gln Lys Phe Lys Gly Lys Ala Lys Leu Thr Ala Val Thr Ser
 85             90             95

Ala Ser Thr Ala Tyr Met Glu Leu Ser Ser Leu Thr Asn Glu Asp Ser
 100            105            110

Ala Val Tyr Tyr Cys Thr Arg Glu Gly Glu Gly Ser Asp Tyr Trp Gly
 115            120            125

Gln Gly Thr Thr Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly
 130            135            140

Gly Gly Ser Gly Gly Gly Gly Ser Gln Ile Val Leu Thr Gln Ser Pro
 145            150            155            160

Ala Thr Met Ala Ala Ser Pro Gly Glu Lys Ile Thr Ile Thr Cys Ser
 165            170            175

Ala Ser Ser Ser Ile Ser Ser Asn Tyr Leu His Trp Tyr Gln Gln Lys
 180            185            190

Pro Gly Phe Ser Pro Lys Leu Leu Ile Tyr Arg Thr Ser Asn Leu Ala
 195            200            205

Ser Gly Val Pro Ala Arg Phe Ser Gly Ser Gly Ser Gly Thr Ser Tyr
 210            215            220

Ser Leu Thr Ile Gly Thr Met Glu Ala Glu Asp Val Ala Thr Tyr Tyr
 225            230            235            240

Cys Gln Gln Gly Ser Ser Ile Pro Tyr Thr Phe Gly Gly Gly Thr Lys
 245            250            255

Leu Glu Leu Ala Ala Ala Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 260            265            270

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
 275            280            285

Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly Val
 290            295            300

His Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Arg Pro
 305            310            315            320

Ser Gln Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Phe Thr Ile Ser
 325            330            335

Ser Gly Tyr Ser Trp His Trp Val Arg Gln Pro Pro Gly Arg Gly Leu
 340            345            350

Glu Trp Ile Gly Tyr Ile Gln Tyr Ser Gly Ile Thr Asn Tyr Asn Pro
 355            360            365

Ser Leu Lys Ser Arg Val Thr Met Leu Val Asp Thr Ser Lys Asn Gln
 370            375            380

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Phe 385	Ser	Leu	Arg	Leu	Ser	Ser	Val	Thr	Ala	Ala	Asp	Thr	Ala	Val	Tyr	
					390						395					
Tyr	Cys	Ala	Arg	Glu 405	Asp	Tyr	Asp	Tyr	His 410	Trp	Tyr	Phe	Asp	Val 415	Trp	
Gly	Gln	Gly	Thr	Thr	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	
					420						425					
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Asp	Ile	Gln	Leu	Thr	Gln	Ser	
					435						440					
Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly	Asp	Arg	Val	Thr	Ile	Thr	Cys	
					450						455					
Ser	Thr	Ser	Ser	Ser	Val	Ser	Tyr	Met	His	Trp	Tyr	Gln	Gln	Lys	Pro	
					465						470					
Gly	Lys	Ala	Pro	Lys	Leu	Leu	Ile	Tyr	Ser	Thr	Ser	Asn	Leu	Ala	Ser	
					485						490					
Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	
					500						505					
Phe	Thr	Ile	Ser	Ser	Leu	Gln	Pro	Glu	Asp	Ile	Ala	Thr	Tyr	Tyr	Cys	
					515						520					
His	Gln	Trp	Ser	Ser	Tyr	Pro	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	
					530						535					
Ile	Lys	Val	Asp	Pro	Ala	Glu	Pro	Lys	Ser	Pro	Asp	Lys	Thr	His	Thr	
					545						550					
Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	
					565						570					
Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ala	Arg	Thr	Pro	Glu	
					580						585					
Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	
					595						600					
Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	
					610						615					
Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	
					625						630					
Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	
					645						650					
Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	
					660						665					
Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	
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Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	
					690						695					
Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	
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Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	
					725						730					
Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	
					740						745					
Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	
					755						760					
His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Lys	Asp	Pro	
					770						775					
Lys	Phe	Trp	Val	Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	
					785						790					
Leu	Leu	Val	Thr	Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Arg	Ser	Lys	Arg	

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805								810				815				
Ser	Arg	Leu	Leu	His	Ser	Asp	Tyr	Met	Asn	Met	Thr	Pro	Arg	Arg	Pro	
820								825				830				
Gly	Pro	Thr	Arg	Lys	His	Tyr	Gln	Ala	Tyr	Ala	Ala	Ala	Arg	Asp	Phe	
835								840				845				
Ala	Ala	Tyr	Arg	Ser	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	
850								855				860				
Pro	Ala	Tyr	Gln	Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	
865								870				875				
Gly	Arg	Arg	Glu	Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	
885								890				895				
Pro	Glu	Met	Gly	Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	
900								905				910				
Tyr	Asn	Glu	Leu	Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	
915								920				925				
Gly	Met	Lys	Gly	Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	
930								935				940				
Gln	Gly	Leu	Ser	Thr	Ala	Thr	Lys	Asp	Thr	Tyr	Asp	Ala	Leu	His	Met	
945								950				955				
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<210> SEQ ID NO 19																
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Met	Asp	Phe	Gln	Val	Gln	Ile	Phe	Ser	Phe	Leu	Leu	Ile	Ser	Ala	Ser	
1			5						10					15		
gtc	ata	atg	tct	aga	atg	gcc	cag	gtg	caa	ctg	cag	cag	tca	ggg	gct	96
Val	Ile	Met	Ser	Arg	Met	Ala	Gln	Val	Gln	Leu	Gln	Gln	Ser	Gly	Ala	
			20					25					30			
gag	ctg	gct	aga	cct	ggg	gct	tca	gtg	aag	atg	tcc	tgc	aag	gct	tct	144
Glu	Leu	Ala	Arg	Pro	Gly	Ala	Ser	Val	Lys	Met	Ser	Cys	Lys	Ala	Ser	
		35				40						45				
ggc	tac	acc	ttt	act	acc	tac	aca	ata	cac	tgg	gta	aga	cgg	agg	cct	192
Gly	Tyr	Thr	Phe	Thr	Thr	Tyr	Thr	Ile	His	Trp	Val	Arg	Arg	Arg	Pro	
	50					55					60					
gga	cac	gat	ctg	gaa	tgg	att	gga	tac	att	aat	cct	agc	agt	gga	tgt	240
Gly	His	Asp	Leu	Glu	Trp	Ile	Gly	Tyr	Ile	Asn	Pro	Ser	Ser	Gly	Cys	
65				70					75					80		
tct	gac	tac	aat	caa	aac	ttc	aag	ggc	aag	acc	aca	ttg	act	gca	gac	288
Ser	Asp	Tyr	Asn	Gln	Asn	Phe	Lys	Gly	Lys	Thr	Thr	Leu	Thr	Ala	Asp	
			85					90						95		
aag	tcc	tcc	aac	aca	gcc	tac	atg	caa	ctg	aac	agc	ctg	aca	tct	gag	336
Lys	Ser	Ser	Asn	Thr	Ala	Tyr	Met	Gln	Leu	Asn	Ser	Leu	Thr	Ser	Glu	
			100					105					110			
gac	tct	gcg	gtc	tat	tac	tgt	gca	aga	aga	gcg	gac	tat	ggt	aac	tac	384
Asp	Ser	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Arg	Ala	Asp	Tyr	Gly	Asn	Tyr	
			115				120					125				
gaa	tat	acc	tgg	ttt	gct	tac	tgg	ggc	caa	ggg	acc	acg	gtc	acc	gtc	432
Glu	Tyr	Thr	Trp	Phe	Ala	Tyr	Trp	Gly	Gln	Gly	Thr	Thr	Val	Thr	Val	

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130	135	140	
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tcg gtc atc gag ctc act Ser Val Ile Glu Leu Thr 165	cag tct cca aaa ttc Gln Ser Pro Lys Phe 170	atg tcc aca tca gta Met Ser Thr Ser Val 175	528
gga gac agg gtc aac gtc Gly Asp Arg Val Asn Val 180	acc tac aag gcc agt Thr Tyr Lys Ala Ser 185	cag aat gtg ggt act Gln Asn Val Gly Thr 190	576
aat gta gcc tgg ttt caa Asn Val Ala Trp Phe 195	caa aaa cca ggg caa Gln Lys Pro Gly Gln 200	tct cct aaa gtt ctg Ser Pro Lys Val Leu 205	624
att tac tcg gca tct tac Ile Tyr Ser Ala Ser 210	cga tac agt gga gtc Arg Tyr Ser Gly Val 215	cct gat cgc ttc aca Pro Asp Arg Phe Thr 220	672
ggc agt gga tct gga aca Gly Ser Gly Ser Gly 225	gat ttc act ctc acc Thr Asp Phe Thr Leu 230	atc agc aat gtg cag Ile Ser Asn Val Gln 235	720
tct gaa gac ttg gca gag Ser Glu Asp Leu Ala 245	tat ttc tgt cag caa Tyr Phe Cys Gln Gln 250	tat cac acc tat cct Tyr His Thr Tyr Pro 255	768
ctc acg ttc gga ggg ggc Leu Thr Phe Gly Gly 260	acc aag ctg gaa atc Thr Lys Leu Glu Ile 265	aaa cgg gcg gcc gct Lys Arg Ala Ala Ala 270	816
ggg ggc gga tct gga gga Gly Gly Gly Gly Ser 275	ggg ggt ggt ggt tcc Gly Gly Gly Ser Gly 280	gga gga ggt ggt tca Gly Gly Gly Gly Ser 285	864
ggg ggc ggc tcc gga gga Gly Gly Gly Ser Gly 290	ggg gga tca gga ggc Gly Gly Ser Gly Gly 295	ggg ggc agc ggc ggc Gly Gly Ser Gly Gly 300	912
gga gga tct ggt ggc ggt Gly Gly Ser Gly Gly 305	ggg ggc tcc gtc gac Gly Gly Ser Val Asp 310	atg gcg cag gtc cag Met Ala Gln Val Gln 315	960
cag gag tcc ggc gga ggc Gln Glu Ser Gly Gly 325	ctg gtg cag cca ggc Leu Val Gln Pro Gly 330	ggg tcc ctg aag ctg Gly Ser Leu Lys Leu 335	1008
agc tgt gct gca tcc ggt Ser Cys Ala Ala Ser 340	ttt acc ttt tcc agt Phe Thr Ser Ser Tyr 345	tac gga atg tcc tgg Gly Met Ser Ser Trp 350	1056
gtt cgc cag acc ccc gat Val Arg Gln Thr Pro 355	aag cgg ctg gaa ctg Lys Arg Leu Glu Leu 360	gtg gcg aca atc aac Val Ala Thr Ile Asn 365	1104
tcc aat ggc gga tct act Ser Asn Gly Gly Ser 370	tat tac cca gat tct Tyr Tyr Pro Asp Ser 375	gtt aaa ggc cgt ttt Val Lys Gly Arg Phe 380	1152
acc atc tcc agg gat aac Thr Ile Ser Arg Asp 385	gcc aaa aac acc ctg Asn Ala Lys Asn Thr 390	tat ctg cag atg tct Leu Tyr Leu Gln Met 395	1200
tct ttg aag agt gag gac Ser Leu Lys Ser Glu 405	act gct atg tac tat Thr Ala Met Tyr Tyr 410	tgc gcc cgg gat cgt Cys Ala Arg Asp Arg 415	1248
gac ggc tac gat gaa ggt Asp Gly Tyr Asp Glu 420	ttc gac tac tgg gga Phe Asp Tyr Trp Gly 425	cag ggc aca acc gtc Gln Gly Thr Thr Val 430	1296
act gtc tct tcc ggc ggt Thr Val Ser Ser Gly 435	ggg ggc agc ggc ggc Gly Gly Ser Gly Gly 440	ggg gga agc ggc ggc Gly Gly Ser Gly Gly 445	1344
ggg ggt tcc gac atc gag Gly Thr Ser Ser Thr 450	ctc acc caa tcc ccc Gly Thr Ser Ser Thr 455	tcc atc atg tcc gcc Gly Thr Ser Ser Thr 460	1392

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Gly	Gly	Ser	Asp	Ile	Glu	Leu	Thr	Gln	Ser	Pro	Ser	Ile	Met	Ser	Ala	
450						455				460						
tcc	cca	ggc	gag	aag	gtc	acc	atg	acc	tgc	tct	gct	tct	tcc	tcc	gtg	1440
Ser	Pro	Gly	Glu	Lys	Val	Thr	Met	Thr	Cys	Ser	Ala	Ser	Ser	Ser	Val	
465					470				475						480	
tct	tac	atg	cat	tgg	tac	cag	cag	aaa	tct	ggt	acc	tct	ccg	aaa	cgc	1488
Ser	Tyr	Met	His	Trp	Tyr	Gln	Gln	Lys	Ser	Gly	Thr	Ser	Pro	Lys	Arg	
				485					490					495		
tgg	atc	tat	gac	acc	tct	aag	ttg	gct	tcc	ggg	gtg	cca	gcc	cgg	ttc	1536
Trp	Ile	Tyr	Asp	Thr	Ser	Lys	Leu	Ala	Ser	Gly	Val	Pro	Ala	Arg	Phe	
				500				505					510			
tct	ggc	agc	gga	agt	gga	acc	tcc	tac	tcc	ctc	act	att	tct	tct	atg	1584
Ser	Gly	Ser	Gly	Ser	Gly	Thr	Ser	Tyr	Ser	Leu	Thr	Ile	Ser	Ser	Met	
			515					520				525				
gag	gct	gaa	gac	gcc	gcg	acc	tat	tat	tgt	cag	caa	tgg	agc	agc	aac	1632
Glu	Ala	Glu	Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Trp	Ser	Ser	Asn	
				530				535				540				
ccc	cct	act	ctg	ggg	cgc	cgt	act	cag	ctg	cag	ctg	aaa	cgc	gcg	gat	1680
Pro	Pro	Thr	Leu	Gly	Arg	Arg	Thr	Gln	Leu	Gln	Leu	Lys	Arg	Ala	Asp	
545					550					555					560	
ccc	gcc	gag	ccc	aaa	tct	cct	gac	aaa	act	cac	aca	tgc	cca	ccg	tgc	1728
Pro	Ala	Glu	Pro	Lys	Ser	Pro	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	
				565					570					575		
cca	gca	cct	cca	gtc	gcg	gga	ccg	tca	gtc	ttc	ctc	ttc	ccc	cca	aaa	1776
Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	
				580					585					590		
ccc	aag	gac	acc	ctc	atg	atc	gcc	cgg	acc	cct	gag	gtc	aca	tgc	gtg	1824
Pro	Lys	Asp	Thr	Leu	Met	Ile	Ala	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	
			595					600					605			
gtg	gtg	gac	gtg	agc	cac	gaa	gac	cct	gag	gtc	aag	ttc	aac	tgg	tac	1872
Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	
			610					615				620				
gtg	gac	ggc	gtg	gag	gtg	cat	aat	gcc	aag	aca	aag	ccg	cgg	gag	gag	1920
Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	
					630					635					640	
cag	tac	aac	agc	acg	tac	cgt	gtg	gtc	agc	gtc	ctc	acc	gtc	ctg	cac	1968
Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	
				645					650					655		
cag	gac	tgg	ctg	aat	ggc	aag	gag	tac	aag	tgc	aag	gtc	tcc	aac	aaa	2016
Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	
				660					665				670			
gcc	ctc	cca	gcc	ccc	atc	gag	aaa	acc	atc	tcc	aaa	gcc	aaa	ggg	cag	2064
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	
				675				680					685			
ccc	cga	gaa	cca	cag	gtg	tac	acc	ctg	ccc	cca	tcc	cgg	gat	gag	ctg	2112
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	
				690				695				700				
acc	aag	aac	cag	gtc	agc	ctg	acc	tgc	ctg	gtc	aaa	ggc	ttc	tat	ccc	2160
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	
					710					715					720	
agc	gac	atc	gcc	gtg	gag	tgg	gag	agc	aat	ggg	cag	ccg	gag	aac	aac	2208
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	
				725					730					735		
tac	aag	acc	acg	cct	ccc	gtg	ctg	gac	tcc	gac	ggc	tcc	ttc	ttc	ctc	2256
Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	
				740				745					750			
tac	agc	aag	ctc	acc	gtg	gac	aag	agc	agg	tgg	cag	cag	ggg	aac	gtc	2304
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	
				755				760					765			

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ttc tca tgc tcc gtg atg cat gag gct ctg cac aac cac tac acg cag	2352
Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln	
770 775 780	
aag agc ctc tcc ctg tct ccg ggt aaa aaa gat ccc aaa ttt tgg gtg	2400
Lys Ser Leu Ser Leu Ser Pro Gly Lys Lys Asp Pro Lys Phe Trp Val	
785 790 795 800	
ctg gtg gtg gtt ggt gga gtc ctg gct tgc tat agc ttg cta gta aca	2448
Leu Val Val Val Gly Gly Val Leu Ala Cys Tyr Ser Leu Leu Val Thr	
805 810 815	
gtg gcc ttt att att ttc tgg gtg agg agt aag agg agc agg ctc ctg	2496
Val Ala Phe Ile Ile Phe Trp Val Arg Ser Lys Arg Ser Arg Leu Leu	
820 825 830	
cac agt gac tac atg aac atg act ccc cgc cgc ccc ggg ccc acc cgc	2544
His Ser Asp Tyr Met Asn Met Thr Pro Arg Arg Pro Gly Pro Thr Arg	
835 840 845	
aag cat tac cag gcc tat gcc gcc gca cgc gac ttc gca gcc tat cgc	2592
Lys His Tyr Gln Ala Tyr Ala Ala Ala Arg Asp Phe Ala Ala Tyr Arg	
850 855 860	
tcc ctg aga gtg aag ttc agc agg agc gca gac gcc ccc gcg tac cag	2640
Ser Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln	
865 870 875 880	
cag gcc cag aac cag ctc tat aac gag ctc aat cta gga cga aga gag	2688
Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu	
885 890 895	
gag tac gat gtt ttg gac aag aga cgt gcc cgg gac cct gag atg ggg	2736
Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly	
900 905 910	
gga aag ccg aga agg aag aac cct cag gaa gcc ctg tac aat gaa ctg	2784
Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu	
915 920 925	
cag aaa gat aag atg gcg gag gcc tac agt gag att ggg atg aaa gcc	2832
Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly	
930 935 940	
gag cgc cgg agg gcc aag ggg cac gat gcc ctt tac cag ggt ctc agt	2880
Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser	
945 950 955 960	
aca gcc acc aag gac acc tac gac gcc ctt cac atg cag gcc ctg ccc	2928
Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro	
965 970 975	
cct cgc taa	2937
Pro Arg	

<210> SEQ ID NO 20

<211> LENGTH: 978

<212> TYPE: PRT

<213> ORGANISM: artificial

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 20

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Val Ile Met Ser Arg Met Ala Gln Val Gln Leu Gln Gln Ser Gly Ala	
20 25 30	
Glu Leu Ala Arg Pro Gly Ala Ser Val Lys Met Ser Cys Lys Ala Ser	
35 40 45	
Gly Tyr Thr Phe Thr Thr Tyr Thr Ile His Trp Val Arg Arg Arg Pro	
50 55 60	
Gly His Asp Leu Glu Trp Ile Gly Tyr Ile Asn Pro Ser Ser Gly Cys	
65 70 75 80	

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Ser	Asp	Tyr	Asn	Gln	Asn	Phe	Lys	Gly	Lys	Thr	Thr	Leu	Thr	Ala	Asp	
			85						90					95		
Lys	Ser	Ser	Asn	Thr	Ala	Tyr	Met	Gln	Leu	Asn	Ser	Leu	Thr	Ser	Glu	
			100					105					110			
Asp	Ser	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Arg	Ala	Asp	Tyr	Gly	Asn	Tyr	
		115					120					125				
Glu	Tyr	Thr	Trp	Phe	Ala	Tyr	Trp	Gly	Gln	Gly	Thr	Thr	Val	Thr	Val	
	130					135					140					
Ser	Ser	Ser	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	
145				150					155					160		
Ser	Val	Ile	Glu	Leu	Thr	Gln	Ser	Pro	Lys	Phe	Met	Ser	Thr	Ser	Val	
			165						170					175		
Gly	Asp	Arg	Val	Asn	Val	Thr	Tyr	Lys	Ala	Ser	Gln	Asn	Val	Gly	Thr	
			180					185					190			
Asn	Val	Ala	Trp	Phe	Gln	Gln	Lys	Pro	Gly	Gln	Ser	Pro	Lys	Val	Leu	
		195					200					205				
Ile	Tyr	Ser	Ala	Ser	Tyr	Arg	Tyr	Ser	Gly	Val	Pro	Asp	Arg	Phe	Thr	
	210					215					220					
Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Asn	Val	Gln	
225					230					235				240		
Ser	Glu	Asp	Leu	Ala	Glu	Tyr	Phe	Cys	Gln	Gln	Tyr	His	Thr	Tyr	Pro	
			245					250						255		
Leu	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Arg	Ala	Ala	Ala	
		260						265					270			
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	
	275					280					285					
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	
	290				295					300						
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Val	Asp	Met	Ala	Gln	Val	Gln	Leu	
305					310					315					320	
Gln	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Gly	Ser	Leu	Lys	Leu	
			325					330						335		
Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe	Ser	Ser	Tyr	Gly	Met	Ser	Trp	
		340						345					350			
Val	Arg	Gln	Thr	Pro	Asp	Lys	Arg	Leu	Glu	Leu	Val	Ala	Thr	Ile	Asn	
		355				360						365				
Ser	Asn	Gly	Gly	Ser	Thr	Tyr	Tyr	Pro	Asp	Ser	Val	Lys	Gly	Arg	Phe	
	370					375					380					
Thr	Ile	Ser	Arg	Asp	Asn	Ala	Lys	Asn	Thr	Leu	Tyr	Leu	Gln	Met	Ser	
385					390					395				400		
Ser	Leu	Lys	Ser	Glu	Asp	Thr	Ala	Met	Tyr	Tyr	Cys	Ala	Arg	Asp	Arg	
			405						410					415		
Asp	Gly	Tyr	Asp	Glu	Gly	Phe	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Thr	Val	
		420						425					430			
Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	
	435					440					445					
Gly	Gly	Ser	Asp	Ile	Glu	Leu	Thr	Gln	Ser	Pro	Ser	Ile	Met	Ser	Ala	
	450					455					460					
Ser	Pro	Gly	Glu	Lys	Val	Thr	Met	Thr	Cys	Ser	Ala	Ser	Ser	Ser	Val	
465					470					475					480	
Ser	Tyr	Met	His	Trp	Tyr	Gln	Gln	Lys	Ser	Gly	Thr	Ser	Pro	Lys	Arg	
			485					490						495		

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Trp	Ile	Tyr	Asp	Thr	Ser	Lys	Leu	Ala	Ser	Gly	Val	Pro	Ala	Arg	Phe
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Ser	Gly	Ser	Gly	Ser	Gly	Thr	Ser	Tyr	Ser	Leu	Thr	Ile	Ser	Ser	Met
		515					520					525			
Glu	Ala	Glu	Asp	Ala	Ala	Thr	Tyr	Tyr	Cys	Gln	Gln	Trp	Ser	Ser	Asn
	530					535					540				
Pro	Pro	Thr	Leu	Gly	Arg	Arg	Thr	Gln	Leu	Gln	Leu	Lys	Arg	Ala	Asp
545					550					555					560
Pro	Ala	Glu	Pro	Lys	Ser	Pro	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys
				565					570					575	
Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys
			580					585					590		
Pro	Lys	Asp	Thr	Leu	Met	Ile	Ala	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val
		595					600					605			
Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr
	610					615					620				
Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu
625					630					635					640
Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His
				645					650					655	
Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys
			660					665					670		
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln
		675					680					685			
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu
	690					695					700				
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro
705				710						715					720
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn
			725						730					735	
Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu
			740					745					750		
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val
		755					760					765			
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln
	770					775					780				
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Lys	Asp	Pro	Lys	Phe	Trp	Val
785					790					795					800
Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	Leu	Val	Thr
			805						810					815	
Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Arg	Ser	Lys	Arg	Ser	Arg	Leu	Leu
			820					825					830		
His	Ser	Asp	Tyr	Met	Asn	Met	Thr	Pro	Arg	Arg	Pro	Gly	Pro	Thr	Arg
		835					840					845			
Lys	His	Tyr	Gln	Ala	Tyr	Ala	Ala	Ala	Arg	Asp	Phe	Ala	Ala	Tyr	Arg
	850					855					860				
Ser	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln
865					870					875					880
Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu
			885					890					895		
Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly
		900						905					910		
Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu

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915	920	925	
Gln Lys Asp Lys Met Ala	Glu Ala Tyr Ser Glu	Ile Gly Met Lys Gly	
930	935	940	
Glu Arg Arg Arg Gly Lys	Gly His Asp Gly Leu	Tyr Gln Gly Leu Ser	
945	950	955	960
Thr Ala Thr Lys Asp Thr	Tyr Asp Ala Leu His	Met Gln Ala Leu Pro	
	965	970	975
Pro Arg			
<210> SEQ ID NO 21			
<211> LENGTH: 3006			
<212> TYPE: DNA			
<213> ORGANISM: artificial			
<220> FEATURE:			
<223> OTHER INFORMATION: synthetic molecule			
<220> FEATURE:			
<221> NAME/KEY: CDS			
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Met Asp Phe Gln Val Gln Ile Phe Ser Phe Leu Leu Ile Ser Ala Ser			
1 5 10 15			
gtc ata atg tct aga atg gcc cag gtg caa ctg cag cag tca ggg gct			96
Val Ile Met Ser Arg Met Ala Gln Val Gln Leu Gln Gln Ser Gly Ala			
20 25 30			
gag ctg gct aga cct ggg gct tca gtg aag atg tcc tgc aag gct tct			144
Glu Leu Ala Arg Pro Gly Ala Ser Val Lys Met Ser Cys Lys Ala Ser			
35 40 45			
ggc tac acc ttt act acc tac aca ata cac tgg gta aga cgg agg cct			192
Gly Tyr Thr Phe Thr Thr Tyr Thr Ile His Trp Val Arg Arg Arg Pro			
50 55 60			
gga cac gat ctg gaa tgg att gga tac att aat cct agc agt gga tgt			240
Gly His Asp Leu Glu Trp Ile Gly Tyr Ile Asn Pro Ser Ser Gly Cys			
65 70 75 80			
tct gac tac aat caa aac ttc aag ggc aag acc aca ttg act gca gac			288
Ser Asp Tyr Asn Gln Asn Phe Lys Gly Lys Thr Thr Leu Thr Ala Asp			
85 90 95			
aag tcc tcc aac aca gcc tac atg caa ctg aac agc ctg aca tct gag			336
Lys Ser Ser Asn Thr Ala Tyr Met Gln Leu Asn Ser Leu Thr Ser Glu			
100 105 110			
gac tct gcg gtc tat tac tgt gca aga aga gcg gac tat ggt aac tac			384
Asp Ser Ala Val Tyr Tyr Cys Ala Arg Arg Ala Asp Tyr Gly Asn Tyr			
115 120 125			
gaa tat acc tgg ttt gct tac tgg ggc caa ggg acc acg gtc acc gtc			432
Glu Tyr Thr Trp Phe Ala Tyr Trp Gly Gln Gly Thr Thr Val Thr Val			
130 135 140			
tcc tca agt gga ggc ggt tca ggt gga ggt ggc tct ggc ggt ggc gga			480
Ser Ser Ser Gly Gly Gly Ser Gly Gly Gly Ser Gly Gly Gly Gly			
145 150 155 160			
tcg gtc atc gag ctc act cag tct cca aaa ttc atg tcc aca tca gta			528
Ser Val Ile Glu Leu Thr Gln Ser Pro Lys Phe Met Ser Thr Ser Val			
165 170 175			
gga gac agg gtc aac gtc acc tac aag gcc agt cag aat gtg ggt act			576
Gly Asp Arg Val Asn Val Thr Tyr Lys Ala Ser Gln Asn Val Gly Thr			
180 185 190			
aat gta gcc tgg ttt caa caa aaa cca ggg caa tct cct aaa gtt ctg			624
Asn Val Ala Trp Phe Gln Gln Lys Pro Gly Gln Ser Pro Lys Val Leu			
195 200 205			
att tac tcg gca tct tac cga tac agt gga gtc cct gat cgc ttc aca			672

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Ile	Tyr	Ser	Ala	Ser	Tyr	Arg	Tyr	Ser	Gly	Val	Pro	Asp	Arg	Phe	Thr	
210					215					220						
ggc	agt	gga	tct	gga	aca	gat	ttc	act	ctc	acc	atc	agc	aat	gtg	cag	720
Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr	Ile	Ser	Asn	Val	Gln	
225					230					235					240	
tct	gaa	gac	ttg	gca	gag	tat	ttc	tgt	cag	caa	tat	cac	acc	tat	cct	768
Ser	Glu	Asp	Leu	Ala	Glu	Tyr	Phe	Cys	Gln	Gln	Tyr	His	Thr	Tyr	Pro	
			245						250					255		
ctc	acg	ttc	gga	ggg	ggc	acc	aag	ctg	gaa	atc	aaa	cgg	gcg	gcc	gca	816
Leu	Thr	Phe	Gly	Gly	Gly	Thr	Lys	Leu	Glu	Ile	Lys	Arg	Ala	Ala	Ala	
			260					265					270			
gga	ggc	ggg	ggg	ggt	tct	ggg	ggc	gga	agc	ggg	ggc	ggg	ggg	tca	gga	864
Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	
			275					280					285			
ggc	ggg	ggg	tcc	ggg	ggg	ggc	ggc	agt	ggg	ggg	ggc	agg	ggg	ggg	ggg	912
Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	
			290					295					300			
ggc	ggc	agt	ggg	ggg	ggc	ggc	agt	gtc	gac	ggg	gtc	cac	tcc	cag	gtg	960
Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Val	Asp	Gly	Val	His	Ser	Gln	Val	
			305			310				315					320	
cag	att	ttc	agc	ttc	ctg	cta	atc	agt	gcc	tca	gtc	ata	atg	tct	aga	1008
Gln	Ile	Phe	Ser	Phe	Leu	Leu	Ile	Ser	Ala	Ser	Val	Ile	Met	Ser	Arg	
				325					330					335		
atg	gcc	cag	gtg	aag	ctg	cag	cag	tct	ggc	gct	gag	ttg	gtg	aaa	cct	1056
Met	Ala	Gln	Val	Lys	Leu	Gln	Gln	Ser	Gly	Ala	Glu	Leu	Val	Lys	Pro	
			340					345					350			
ggg	gct	tca	gtg	aag	ata	tcc	tgc	aag	gct	tct	ggc	tac	acc	ttc	act	1104
Gly	Ala	Ser	Val	Lys	Ile	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Thr	
			355					360					365			
gac	cat	gct	att	cac	tgg	gcg	aag	cag	aag	cct	gaa	cag	ggc	ctg	gaa	1152
Asp	His	Ala	Ile	His	Trp	Ala	Lys	Gln	Lys	Pro	Glu	Gln	Gly	Leu	Glu	
			370			375					380					
tgg	att	gga	tat	att	tct	ccc	gga	aat	gat	gat	att	aag	tac	aat	gag	1200
Trp	Ile	Gly	Tyr	Ile	Ser	Pro	Gly	Asn	Asp	Asp	Ile	Lys	Tyr	Asn	Glu	
				385		390				395					400	
aag	ttc	aag	ggc	aag	gcc	aca	ctg	act	gca	gac	aaa	tcc	tcc	agc	act	1248
Lys	Phe	Lys	Gly	Lys	Ala	Thr	Leu	Thr	Ala	Asp	Lys	Ser	Ser	Ser	Thr	
				405					410					415		
gcc	tac	atg	cag	ctc	aac	agc	ctg	aca	tct	gag	gat	tct	gca	gtg	tat	1296
Ala	Tyr	Met	Gln	Leu	Asn	Ser	Leu	Thr	Ser	Glu	Asp	Ser	Ala	Val	Tyr	
				420				425					430			
ttc	tgt	aaa	aga	tcg	tac	tac	ggc	cac	tgg	ggc	caa	ggc	acc	acg	gtc	1344
Phe	Cys	Lys	Arg	Ser	Tyr	Tyr	Gly	His	Trp	Gly	Gln	Gly	Thr	Thr	Val	
			435				440					445				
acc	gtc	tcc	tca	ggg	gga	ggc	ggg	tca	ggc	gga	ggg	ggc	tct	ggc	ggg	1392
Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	
			450			455					460					
ggc	gga	tcg	gac	atc	gag	ctc	act	cag	tct	cca	gcc	tcc	tta	tct	gta	1440
Gly	Gly	Ser	Asp	Ile	Glu	Leu	Thr	Gln	Ser	Pro	Ala	Ser	Leu	Ser	Val	
				465		470				475					480	
tct	gtg	gga	gaa	act	gtc	acc	atc	aca	tgt	cga	gca	agt	gag	aat	att	1488
Ser	Val	Gly	Glu	Thr	Val	Thr	Ile	Thr	Cys	Arg	Ala	Ser	Glu	Asn	Ile	
				485				490						495		
tac	agt	aat	tta	gca	tgg	tac	caa	cag	aaa	cag	gga	aaa	tct	cct	cag	1536
Tyr	Ser	Asn	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Gln	Gly	Lys	Ser	Pro	Gln	
			500					505					510			
ctc	ctg	gtc	tat	gct	gca	aca	aac	tta	gca	gat	ggg	gtg	cca	tca	agg	1584
Leu	Leu	Val	Tyr	Ala	Ala	Thr	Asn	Leu	Ala	Asp	Gly	Val	Pro	Ser	Arg	
			515					520					525			

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ttc agt ggc agt gga tgc ggc aca cag tat tcc ctc aag atc aac agc	1632
Phe Ser Gly Ser Gly Ser Gly Thr Gln Tyr Ser Leu Lys Ile Asn Ser	
530 535 540	
ctg cag tct gaa gat ttt ggg agt tat tac tgt caa cat ttt tgg ggt	1680
Leu Gln Ser Glu Asp Phe Gly Ser Tyr Tyr Cys Gln His Phe Trp Gly	
545 550 555 560	
act ccg tac acg ttc gga ggg ggg acc aag ctg gag ctg aaa gtg gat	1728
Thr Pro Tyr Thr Phe Gly Gly Gly Thr Lys Leu Glu Leu Lys Val Asp	
565 570 575	
ccc gcc gag ccc aaa tct cct gac aaa act cac aca tgc cca ccg tgc	1776
Pro Ala Glu Pro Lys Ser Pro Asp Lys Thr His Thr Cys Pro Pro Cys	
580 585 590	
cca gca cct cca gtc gcg gga ccg tca gtc ttc ctc ttc ccc cca aaa	1824
Pro Ala Pro Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro Pro Lys	
595 600 605	
ccc aag gac acc ctc atg atc gcc cgg acc cct gag gtc aca tgc gtg	1872
Pro Lys Asp Thr Leu Met Ile Ala Arg Thr Pro Glu Val Thr Cys Val	
610 615 620	
gtg gtg gac gtg agc cac gaa gac cct gag gtc aag ttc aac tgg tac	1920
Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr	
625 630 635 640	
gtg gac ggc gtg gag gtg cat aat gcc aag aca aag ccg cgg gag gag	1968
Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu	
645 650 655	
cag tac aac agc acg tac cgt gtg gtc agc gtc ctc acc gtc ctg cac	2016
Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His	
660 665 670	
cag gac tgg ctg aat ggc aag gag tac aag tgc aag gtc tcc aac aaa	2064
Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys	
675 680 685	
gcc ctc cca gcc ccc atc gag aaa acc atc tcc aaa gcc aaa ggg cag	2112
Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln	
690 695 700	
ccc cga gaa cca cag gtg tac acc ctg ccc cca tcc cgg gat gag ctg	2160
Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu	
705 710 715 720	
acc aag aac cag gtc agc ctg acc tgc ctg gtc aaa ggc ttc tat ccc	2208
Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro	
725 730 735	
agc gac atc gcc gtg gag tgg gag agc aat ggg cag ccg gag aac aac	2256
Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn	
740 745 750	
tac aag acc acg cct ccc gtg ctg gac tcc gac ggc tcc ttc ttc ctc	2304
Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu	
755 760 765	
tac agc aag ctc acc gtg gac aag agc agg tgg cag cag ggg aac gtc	2352
Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val	
770 775 780	
ttc tca tgc tcc gtg atg cat gag gct ctg cac aac cac tac acg cag	2400
Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln	
785 790 795 800	
aag agc ctc tcc ctg tct ccg ggt aaa aaa gat ccc aaa ttt tgg gtg	2448
Lys Ser Leu Ser Leu Ser Pro Gly Lys Lys Asp Pro Lys Phe Trp Val	
805 810 815	
ctg gtg gtg gtt ggt gga gtc ctg gct tgc tat agc ttg cta gta aca	2496
Leu Val Val Val Gly Gly Val Leu Ala Cys Tyr Ser Leu Leu Val Thr	
820 825 830	
gtg gcc ttt att att ttc tgg gtg agg agt aag agg agc agg ctc ctg	2544
Val Ala Phe Ile Ile Phe Trp Val Arg Ser Lys Arg Ser Arg Leu Leu	
835 840 845	

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cac agt gac tac atg aac atg act ccc cgc cgc ccc ggg ccc acc cgc	2592
His Ser Asp Tyr Met Asn Met Thr Pro Arg Arg Pro Gly Pro Thr Arg	
850 855 860	
aag cat tac cag gcc tat gcc gcc gca cgc gac ttc gca gcc tat cgc	2640
Lys His Tyr Gln Ala Tyr Ala Ala Ala Arg Asp Phe Ala Ala Tyr Arg	
865 870 875 880	
tcc ctg aga gtg aag ttc agc agg agc gca gac gcc ccc gcg tac cag	2688
Ser Leu Arg Val Lys Phe Ser Arg Ser Ala Asp Ala Pro Ala Tyr Gln	
885 890 895	
cag ggc cag aac cag ctc tat aac gag ctc aat cta gga cga aga gag	2736
Gln Gly Gln Asn Gln Leu Tyr Asn Glu Leu Asn Leu Gly Arg Arg Glu	
900 905 910	
gag tac gat gtt ttg gac aag aga cgt gcc cgg gac cct gag atg ggg	2784
Glu Tyr Asp Val Leu Asp Lys Arg Arg Gly Arg Asp Pro Glu Met Gly	
915 920 925	
gga aag ccg aga agg aag aac cct cag gaa gcc ctg tac aat gaa ctg	2832
Gly Lys Pro Arg Arg Lys Asn Pro Gln Glu Gly Leu Tyr Asn Glu Leu	
930 935 940	
cag aaa gat aag atg gcg gag gcc tac agt gag att ggg atg aaa ggc	2880
Gln Lys Asp Lys Met Ala Glu Ala Tyr Ser Glu Ile Gly Met Lys Gly	
945 950 955 960	
gag cgc cgg agg gcc aag ggg cac gat gcc ctt tac cag ggt ctc agt	2928
Glu Arg Arg Arg Gly Lys Gly His Asp Gly Leu Tyr Gln Gly Leu Ser	
965 970 975	
aca gcc acc aag gac acc tac gac gcc ctt cac atg cag gcc ctg ccc	2976
Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro	
980 985 990	
cct cgc taa tcctcgagag atccggatta g	3006
Pro Arg	

<210> SEQ ID NO 22
 <211> LENGTH: 994
 <212> TYPE: PRT
 <213> ORGANISM: artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic Construct
 <400> SEQUENCE: 22

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Val Ile Met Ser Arg Met Ala Gln Val Gln Leu Gln Ser Gly Ala	
20 25 30	
Glu Leu Ala Arg Pro Gly Ala Ser Val Lys Met Ser Cys Lys Ala Ser	
35 40 45	
Gly Tyr Thr Phe Thr Thr Tyr Thr Ile His Trp Val Arg Arg Arg Pro	
50 55 60	
Gly His Asp Leu Glu Trp Ile Gly Tyr Ile Asn Pro Ser Ser Gly Cys	
65 70 75 80	
Ser Asp Tyr Asn Gln Asn Phe Lys Gly Lys Thr Thr Leu Thr Ala Asp	
85 90 95	
Lys Ser Ser Asn Thr Ala Tyr Met Gln Leu Asn Ser Leu Thr Ser Glu	
100 105 110	
Asp Ser Ala Val Tyr Tyr Cys Ala Arg Arg Ala Asp Tyr Gly Asn Tyr	
115 120 125	
Glu Tyr Thr Trp Phe Ala Tyr Trp Gly Gln Gly Thr Thr Val Thr Val	
130 135 140	
Ser Ser Ser Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly	
145 150 155 160	

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Ser Val Ile Glu Leu Thr Gln Ser Pro Lys Phe Met Ser Thr Ser Val	165	170	175
Gly Asp Arg Val Asn Val Thr Tyr Lys Ala Ser Gln Asn Val Gly Thr	180	185	190
Asn Val Ala Trp Phe Gln Gln Lys Pro Gly Gln Ser Pro Lys Val Leu	195	200	205
Ile Tyr Ser Ala Ser Tyr Arg Tyr Ser Gly Val Pro Asp Arg Phe Thr	210	215	220
Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Asn Val Gln	225	230	235
Ser Glu Asp Leu Ala Glu Tyr Phe Cys Gln Gln Tyr His Thr Tyr Pro	245	250	255
Leu Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg Ala Ala Ala	260	265	270
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly	275	280	285
Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly	290	295	300
Gly Gly Ser Gly Gly Gly Gly Ser Val Asp Gly Val His Ser Gln Val	305	310	315
Gln Ile Phe Ser Phe Leu Leu Ile Ser Ala Ser Val Ile Met Ser Arg	325	330	335
Met Ala Gln Val Lys Leu Gln Gln Ser Gly Ala Glu Leu Val Lys Pro	340	345	350
Gly Ala Ser Val Lys Ile Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr	355	360	365
Asp His Ala Ile His Trp Ala Lys Gln Lys Pro Glu Gln Gly Leu Glu	370	375	380
Trp Ile Gly Tyr Ile Ser Pro Gly Asn Asp Asp Ile Lys Tyr Asn Glu	385	390	395
Lys Phe Lys Gly Lys Ala Thr Leu Thr Ala Asp Lys Ser Ser Ser Thr	405	410	415
Ala Tyr Met Gln Leu Asn Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr	420	425	430
Phe Cys Lys Arg Ser Tyr Tyr Gly His Trp Gly Gln Gly Thr Thr Val	435	440	445
Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly	450	455	460
Gly Gly Ser Asp Ile Glu Leu Thr Gln Ser Pro Ala Ser Leu Ser Val	465	470	475
Ser Val Gly Glu Thr Val Thr Ile Thr Cys Arg Ala Ser Glu Asn Ile	485	490	495
Tyr Ser Asn Leu Ala Trp Tyr Gln Gln Lys Gln Gly Lys Ser Pro Gln	500	505	510
Leu Leu Val Tyr Ala Ala Thr Asn Leu Ala Asp Gly Val Pro Ser Arg	515	520	525
Phe Ser Gly Ser Gly Ser Gly Thr Gln Tyr Ser Leu Lys Ile Asn Ser	530	535	540
Leu Gln Ser Glu Asp Phe Gly Ser Tyr Tyr Cys Gln His Phe Trp Gly	545	550	555
Thr Pro Tyr Thr Phe Gly Gly Gly Thr Lys Leu Glu Leu Lys Val Asp	565	570	575

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Pro	Ala	Glu	Pro	Lys	Ser	Pro	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys
			580					585					590		
Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys
		595					600				605				
Pro	Lys	Asp	Thr	Leu	Met	Ile	Ala	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val
	610				615						620				
Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr
	625				630					635					640
Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu
			645						650					655	
Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His
			660					665					670		
Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys
		675					680					685			
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln
	690					695					700				
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu
	705				710					715					720
Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro
			725					730						735	
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn
		740					745					750			
Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu
		755					760					765			
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val
	770					775					780				
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln
	785				790					795					800
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	Lys	Asp	Pro	Lys	Phe	Trp	Val
			805						810					815	
Leu	Val	Val	Val	Gly	Gly	Val	Leu	Ala	Cys	Tyr	Ser	Leu	Leu	Val	Thr
			820					825					830		
Val	Ala	Phe	Ile	Ile	Phe	Trp	Val	Arg	Ser	Lys	Arg	Ser	Arg	Leu	Leu
		835					840					845			
His	Ser	Asp	Tyr	Met	Asn	Met	Thr	Pro	Arg	Arg	Pro	Gly	Pro	Thr	Arg
	850					855					860				
Lys	His	Tyr	Gln	Ala	Tyr	Ala	Ala	Ala	Arg	Asp	Phe	Ala	Ala	Tyr	Arg
	865				870					875					880
Ser	Leu	Arg	Val	Lys	Phe	Ser	Arg	Ser	Ala	Asp	Ala	Pro	Ala	Tyr	Gln
			885						890				895		
Gln	Gly	Gln	Asn	Gln	Leu	Tyr	Asn	Glu	Leu	Asn	Leu	Gly	Arg	Arg	Glu
		900						905					910		
Glu	Tyr	Asp	Val	Leu	Asp	Lys	Arg	Arg	Gly	Arg	Asp	Pro	Glu	Met	Gly
		915					920					925			
Gly	Lys	Pro	Arg	Arg	Lys	Asn	Pro	Gln	Glu	Gly	Leu	Tyr	Asn	Glu	Leu
		930				935					940				
Gln	Lys	Asp	Lys	Met	Ala	Glu	Ala	Tyr	Ser	Glu	Ile	Gly	Met	Lys	Gly
	945				950					955					960
Glu	Arg	Arg	Arg	Gly	Lys	Gly	His	Asp	Gly	Leu	Tyr	Gln	Gly	Leu	Ser
				965				970						975	

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Thr Ala Thr Lys Asp Thr Tyr Asp Ala Leu His Met Gln Ala Leu Pro
 980 985 990

Pro Arg

The invention claimed is:

1. A method of treating CD30 negative cancer comprising the step of administering to a subject in need thereof a nucleic acid molecule comprising a nucleic acid sequence encoding a recombinant polypeptide with at least the following domains starting from the N-terminus to the C-terminus:

first domain containing i) an anti-CD30 single chain antibody unit and ii) an antibody unit specific for an antigen present on the surface of a predetermined target cell, wherein the target cell is a CD30 negative cancer cell;

optionally a spacer domain

a transmembrane domain; and

a cytoplasmatic signalling domain,

wherein the anti-CD30 single chain antibody unit in the first domain is HRS3 scFv of SEQ ID NO. 2 or a homolog thereof having at least 95% identity with SEQ ID NO. 2 binding specifically to CD30, and wherein the nucleic acid sequence is present in a cell, cell line, host cell or a viral vector or other vector.

2. The method of treating cancer according to claim 1 wherein the cell, cell line, or host cell expresses the recombinant polypeptide.

3. The method of treating cancer according to claim 1, wherein the cancer cells of said cancer express the antigen.

4. The method for treating cancer according to claim 1, wherein the antibody unit used for the antigen present on the surface of the predetermined target cells being CD30 negative binds to a tumor-associated antigen selected from the group consisting of carcinoembryonic antigen (CEA), CA19-9, CA72-4 (TAG-72), PSCA, Muc-1, HMW-MAA, p97 melanotransferrin, fetal actelycholin receptor, ErbB2 (Her2/neu), multi-drug-resistance protein (MDR), CD19, CD20, and TOSO.

5. The method of treating cancer according to claim 1, wherein said recombinant polypeptide further comprises a leader sequence being located N-termini to the first domain containing anti-CD30 single chain antibody unit and the antibody unit specific for the antigen present on the surface of a predetermined target cell being CD30 negative.

6. The method of treating cancer according to claim 1, wherein the anti-CD30 single chain antibody unit is the HRS3 scFv peptide of SEQ ID NO. 2.

7. The method of treating cancer according to claim 1, wherein the spacer domain is present and is an IgG1 CH2 CH3 domain of SEQ ID No. 6 or homologs thereof having at least 95% identity herewith.

8. The method of treating cancer according to claim 1, wherein the transmembrane domain is obtained from CD28.

9. The method of treating cancer according to claim 1, wherein the intracellular domain contains a CD3zeta or a Fc-epsilon receptor I-gamma signalling chain or a costimulatory wall.

10. The method of treating cancer according to claim 1, wherein the cell or the host cell is provided, and wherein the cell or host cell is a modified peripheral blood cell.

11. The method of treating cancer according to claim 1, wherein the spacer domain is a mutated IgG1 CH2 CH3 domain according to SEQ ID No. 6.

12. The method of treating cancer according to claim 1, wherein said antigen present on the surface of a predetermined target cell is a tumor-associated antigen.

13. The method of treating cancer according to claim 1, wherein the intracellular domain is the IgE Fc epsilon receptor-I gamma-signalling unit of SEQ ID No. 12 or a homolog thereof having at least 95% identity.

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